

Optimal biomass-based climate change mitigation pathways are spatially heterogeneous due to localized technoeconomic and biophysical constraints

Dominic Woolf^{*1}, Jocelyn Lavallee², Anna Raffeld², Doria Gordon², and Emily Oldfield²

¹Cornell University

²Environmental Defense Fund

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Abstract

The use of biomass for climate change mitigation offers substantial technical potential, but optimal deployment is highly dependent on spatial, technoeconomic, and biophysical constraints. Blanket climate policies frequently fail to capture this heterogeneity. Here we apply a spatially explicit techno-economic and consequential life cycle assessment framework deployed at a 0.1-degree resolution across the United States, Europe, India, and China. We evaluate the economic viability and carbon abatement potential of three distinct biomass utilization pathways: Bioenergy (BES), Bioenergy with Carbon Capture and Storage (BECCS), and Bioenergy with Biochar Carbon Storage (BEBCS). Localized constraints, such as distance to geological CO₂ sinks, baseline grid carbon intensity, and native soil cation exchange capacity, dictate the economically optimal pathway. Traditional BES tends to dominate at low carbon prices, particularly in regions with high wholesale energy prices like Europe, or high grid carbon intensities like India. Biochar (BEBCS) emerges as a highly competitive transition technology at moderate carbon prices, and serves as the optimal long-term pathway in specific agro-ecological zones with degraded soils, such as southern China. Conversely, high capital costs and piecewise long-distance pipeline penalties act as severe barriers to BECCS, which only achieves universal dominance at high carbon prices (e.g., \$150/tCO₂e) where carbon capture revenues finally overwhelm infrastructure penalties. Ultimately, effective climate mitigation requires localized policies that match biomass resources to their most efficient regional end-use.

^{*}Correspondence to d.woolf@cornell.edu

1 INTRODUCTION

Biochar has been widely promoted for climate change mitigation, with co-benefits for soil health and crop production when used as a soil amendment.(1, 2) Globally, biochar has the sustainable technical potential for carbon sequestration and emissions reductions of up to 3.4–6.3 Pg CO₂e. However, the costs, benefits and tradeoffs for biochar vary spatially, creating substantial heterogeneity in its feasibility at a given carbon price. This variability is further affected by the fact that biochar must compete with alternative uses for the limited supply of available waste biomass. Indeed, alternative uses of biomass, such as producing bioenergy with or without carbon capture and sequestration can also help to mitigate climate change and may provide greater climate mitigation, lower costs, and/or greater co-benefits in certain contexts.(3) This depends on a number of different factors, including soil type, cropland productivity, climate, energy fuel mix, feedstock availability, distance to CO₂ sinks for carbon capture and storage, energy prices and food prices. For example, bioenergy production may provide a greater climate benefit in regions that are highly dependent on carbon-intensive energy sources and have high soil fertility, whereas biochar is more favorable where low-fertility soils can benefit from its soil-health improvement.(3)

It is, therefore, critical to understand where biochar could be the best use of resources from a climate mitigation perspective. Currently, most biochar assessments focus on the technical potential, without mapping out the tradeoffs or identifying optimal pathways based on location. While there are some biochar life-cycle assessments that account for alternative biomass uses, they are highly context-specific and difficult to scale up on(4, 5). As such, this project aims to provide a detailed feasibility and tradeoff analysis across four regions (North America, India, China, and Europe) to map where biochar represents the most efficient use of biomass to reduce atmospheric greenhouse gasses (GHGs). These regions represent a diversity of geographic, socio-economic, and agro-ecological conditions for a broad assessment of how constraints, tradeoffs and benefits vary across regions. This work will allow policy-makers and practitioners to make scientifically-informed decisions about which mitigation pathways are best suited for their geographies. It can also help contextualize biochar’s mitigation potential by identifying areas that maximize its climate benefits and realistically support its production and application.

2 RESULTS

2.1 Capital-intensive pathways are highly sensitive to discount rates and carbon prices

Figure 2 to Figure 5 illustrate the core vulnerability of CDR pathways to capital lock-out. Because both BECCS (heavy upfront infrastructure) and BEBCS (deferred agronomic value) are highly sensitive to the cost of capital, high discount rates act as a strong barrier to adoption, forcing a retreat to traditional Bioenergy (BES) without high carbon prices. For example, in the USA, at a 2% social discount rate and a \$30/t carbon price, BEBCS covers a large extent of the US Midwest and South, with BECCS only being viable close to CO₂ sinks. However, stepping up to the 8% market baseline eliminates BEBCS's advantage, replacing it with almost entirely with BES. The value of biochar relies heavily on liming and nutrient substitution benefits that are amortized over a 10-year agronomic impact duration. At an 8% or 15% discount rate, the present value of those future soil benefits is diminished, rendering the upfront thermal penalties of pyrolysis economically unviable at low carbon prices. At a 2% discount rate and intermediate carbon price (around \$100/t), BECCS is nearly universal across the US. At 15%, the BECCS dominance is fractured, with BES reclaiming the geographic margins (e.g., the Northeast and far West), and BEBCS remaining only in localized pockets where soil conditions make it particularly attractive. This occurs because, when capital is expensive, the penalty for long-distance pipelines becomes insurmountable until carbon prices reach the \$150/t range. By the time the carbon price reaches \$150/t, the maps become almost entirely red across all discount rates. At this pricing level, the high CDR capture efficiency overwhelms the high cost of capital, making BECCS the most economically attractive option.

2.2 Biochar provides a mid-price transition technology option and a long-term solution for specific agro-ecological zones

Figure 5 provides a counterpoint to the US results. It shows how soils with strong production constraints that can be alleviated by biochar can create a persistent competitive advantage for biochar, particularly when combined with large distances to CO₂ storage sites. BEBCS performs a central role as a transition technology that peaks at intermediate carbon prices that are high enough to make its carbon sequestration competitive with BES, but not so high that the more efficient CDR of BECCS per unit biomass outcompetes it (Figure 6). Nonetheless, while BECCS ultimately becomes the optimal technology across the entire US spatial domain at high carbon prices, BEBCS remains the most viable pathway in southern China across almost all price and discount rate scenarios. This demonstrates that BEBCS is not merely a transition technology, but the

optimal long-term pathway for specific agro-ecological zones. In Southern China, characterized by highly weathered, acidic soils, the biochar’s liming and nutrient use efficiency benefits provide a substantial economic credit that buffers against high processing costs. The model scales physical yield efficiency gains inversely to the soil’s native Cation Exchange Capacity (CEC), thus granting severe CEC deficits higher premiums. This is confirmed by the SHAP analysis (Figure 7), which identifies soil CEC as the dominant spatial driver in this region.

2.3 Regional policies hinder biochar competitiveness

In India, on the other hand, the mid carbon price advantage of BEBCS is almost entirely eliminated, with BECCS eventually covering most of the map at high carbon prices. Here, fertilizer subsidies hamper BEBCS’s economic viability. Because BEBCS economics relies on improved nutrient use efficiency and lower fertilizer inputs per unit crop yield, cheap subsidized synthetic fertilizers erode biochar’s nutrient substitution value. Despite this, Biochar maintains a strong foothold in the regions in the North East, North West, and central Eastern coastal regions of India where low soil CEC is a major constraint to crop production.

2.4 Regional energy prices and carbon intensity provide a critical constraint on the optimal CDR pathways

At lower carbon prices The optimal technology in Europe is almost universally BES, driven by high energy prices as the dominant driver across Europe. At \$30 / t CO₂, the revenue from maximized power generation in BES outweighs the incentive to capture and sequester carbon, rendering the capital expenditure for BECCS and BEBCS unviable across all discount rates. As carbon prices increase, BECCS begins to dominate across central and northern Europe. BEBCS gains a strong advantage in Southern Europe, and Eastern Scandinavia where various combinations of large distances to CO₂ sinks, already-low carbon intensity of the energy systems, and infertile soils make it more competitive. As in China, we see BEBCS maintain its advantage in these regions, even at very high carbon prices.

The value of bioenergy power also depends on how it is dispatched within the wider power system, which this plant-level analysis represents only approximately. Each plant here processes a fixed annual quantity of biomass, set by its capacity and capacity factor, and BES chooses between base-load and flexible operation on its own net value. In system models that optimise capacity and hourly dispatch together, biomass is a scarce, storable fuel that the system directs to the hours and uses where it is most valuable, such as periods of high residual load, so that its operating regime depends on the generation mix, demand and other sources of flexibility of each grid (6–8). Where such flexibility is scarce, the value of BES, and of BECCS operated

below base load, may therefore be higher than estimated here; where it is abundant, base-load operation, as assumed here, is closer to the system optimum.

2.5 Competing uses of crop residues

The analysis values crop residues against two counterfactuals: retention in the field, whose net soil GHG effect is close to zero (9), and open burning, whose avoided CH₄ and N₂O emissions are credited. A third use, livestock fodder and bedding, is excluded from the supply: the sustainably harvestable residue of (10), which is net of the fraction that must be retained to avoid soil degradation, is reduced by the fraction already used as fodder and bedding (11). This matters most in India, where cereal straws are a major source of fodder, whereas rice straw in the north-western rice–wheat belt is largely burned. Where the fodder fraction is underestimated locally, diverting residues would displace feed that must be replaced from other sources, with associated emissions, land use and costs; the net abatement estimated here is therefore most robust where residues would otherwise be burned or left in the field.

3 DISCUSSION

The deployment of biomass for climate change mitigation cannot be guided by a single universal pathway. Our spatial analysis demonstrates that the economic viability of BES, BECCS, and BEBCS is dictated by a complex interplay of localized biophysical constraints and regional market dynamics.

While BECCS offers the highest theoretical carbon removal potential, its heavy infrastructure requirements and sensitivity to the cost of capital create significant barriers to entry, particularly where biomass is located far from suitable geological sinks. In contrast, biochar (BEBCS) circumvents these transport constraints and establishes a resilient economic niche in regions with degraded, low-CEC soils, where its agronomic co-benefits—such as liming equivalence and nutrient substitution—offset its lower electrical conversion efficiencies. Southern China exemplifies this dynamic, demonstrating that BEBCS is a robust, long-term mitigation strategy for specific agro-ecological zones rather than merely a transition technology.

Furthermore, baseline energy markets shape optimal pathways. In Europe, high wholesale electricity prices establish BES as the dominant pathway at low carbon prices.

Biochar plants could also sell the heat from their surplus pyrolysis vapours and gases rather than generating electricity. In an exploratory scenario in which BEBCS sold all its heat to a year-round user, heat recovery raised the net value of BEBCS in all four regions, because heat earns more than electricity from the same fuel at a much lower capital cost, while abating slightly less, as heat displaces less CO₂ per unit than electricity.

Year-round heat demand, such as an industrial process or a district heating network, is uncommon in residue-producing regions, so heat recovery is an opportunity for co-location at particular sites rather than a basis for regional deployment, and it is not included in the results.

Realizing the climate mitigation potential of global biomass requires moving beyond blanket technological mandates. Policymakers must adopt spatially explicit strategies that target BECCS deployment near geological sinks, incentivize BEBCS in agro-ecological zones suffering from soil degradation, and recognize the enduring role of BES in displacing highly carbon-intensive fossil grids.

4 METHODS

4.1 Objectives and design

The C-SCAPE (Carbon Spatial Cost and Allocation Pathways Evaluator) model (version 1.0; implemented as the R package BiocharAG) evaluates three biomass utilization pathways: Bioenergy (BES), Bioenergy with Carbon Capture and Storage (BECCS), and Bioenergy with Biochar Carbon Storage (BEBCS). For each grid cell and technology, the model calculates a discounted cost-benefit balance, expressed as a levelized annual net value per megagram of dry, ash-free feedstock (the net present value annualized with the capital recovery factor; referred to as NPV below). GHG emissions and removals are converted to 100-year CO₂ equivalents and internalized using an assumed carbon price.

The model was deployed across the coterminous United States, Europe, India, and China. Each region is restricted to its national territory; Europe comprises the countries of geographic Europe within 11°W–40°E and 35–71°N, including Ukraine, Belarus and the part of Türkiye within these bounds, but excluding Russia. All spatial inputs were harmonized to a geographic (WGS84) grid of 0.1° (0.2° for the United States). Area-dependent calculations use cell areas on the ellipsoid, and the biomass collection distances were computed in region-specific equal-area projections (EPSG:5070, EPSG:3035 and Albers equal-area projections for China and India) before being returned to the model grid. The model was implemented in R, using the terra package for spatial analysis ([12](#)).

The following model description outlines the assumptions, equations and parameters used in the model. The default parameter values are provided in a table at the end of the Methods.

4.2 Biomass feedstock sourcing and logistics

4.2.1 Biomass transport costs and emissions

High resolution rasters of available crop residue biomass were derived from the sustainably harvestable crop residues of Karan et al. (10), which exclude the fraction that must be retained to avoid soil degradation, minus the fraction used as livestock fodder and bedding, estimated for each world region from the stover feed use of (11). The supply modelled here therefore does not divert residues from these uses. The average transport distance for biomass was calculated from this using focal statistics based on the annual biomass requirement for the conversion facility. The required annual biomass mass (M_{req}) is determined by the target thermal capacity, the capacity factor, and the biomass lower heating value (LHV). Collection distances were computed for a reference capacity factor of 0.70 and a series of plant sizes; each technology uses the distance for the plant size with the same annual biomass demand at its own capacity factor, interpolated in the square root of plant size. A spatial focal sum algorithm was applied to aggregate the surrounding biomass over a discrete series of expanding radii (from 5 km up to 500 km). The required collection radius was then calculated via quadratic interpolation between these radial bins. All domains were uniformly projected and resampled to the 0.1 degree (~ 10 km) equal-area grid prior to executing the focal sum algorithms to ensure methodological consistency. Assuming a circular catchment geometry, the average straight-line collection distance is approximated as two-thirds of the interpolated bounding radius (i.e. assuming homogeneous distribution within the catchment area). The average straight-line distance was multiplied by a tortuosity factor of 1.3 to calculate the effective road transport distance.(13) Biomass logistical costs were modeled using a fixed material handling and loading cost and a variable trucking cost applied to the effective road distance. The loading and handling cost is 6.27 \$/Mg (4.59 \$ per dry tonne in 2012 USD, escalated to 2024 USD with the US consumer price index) (14). The variable cost of 0.19 \$/Mg/km (0.12 \$/Mg/km for straw and stover in 2006 USD, escalated to 2024 USD with the US consumer price index) is the truck distance-variable cost of (15), which is expressed per kilometre of one-way distance and includes the empty return trip.

Haulage costs vary with terrain and road access, which the flat per-kilometre rate does not capture. For every candidate plant location and plant size, travel from each field in the collection area to the plant was routed as the least-time path on the global motorised friction surface of (16), at 30 arc-seconds. Biomass-weighted over the collection area, this gives the travel time and road-path length per kilometre of straight-line distance, and the cumulative climb on the loaded and empty legs. The variable trucking cost was divided into time-based costs (driver wages and benefits, truck finance, insurance; 64%), fuel (24%) and other distance-based costs (repairs, tyres, tolls; 12%), following (17). The time-based share scales with travel time, fuel with road

distance and climb, and the remainder with road distance. Climb fuel is the energy to lift the loaded and empty truck over the cumulative ascent of each leg; descents are not credited, because diesel trucks recover little of the energy on downhill grades. Each factor is normalised to a biomass-weighted mean of one within each region, because the regional level of haulage cost is set by the regional haulage cost factor (Table 1). The factors therefore redistribute haulage cost within a region, from flat, well-connected areas to rugged or poorly connected ones. Biochar is returned to the fields as a backhaul in the feedstock trucks, which run year-round from field-edge storage, so only its loading and handling cost is charged. Field application (spreading and incorporation) is costed per hectare at the biochar application rate (18), and the diesel used by the tractor (17.5 L ha⁻¹, the median of published estimates for spreading and incorporation (19–22)) is counted as an emission. Scope three transportation emissions associated with feedstock procurement were estimated using a heavy-duty diesel transport emissions factor applied to the effective distance, scaled with the fuel factors, and deducted from the net carbon abatement.

4.2.2 Fuel quality

Agricultural crop residues typically have a higher ash content than the wood chips more commonly used in bioenergy combustion. High-ash feedstocks carry significant fouling and slagging risks, requiring additional capital expenditures (CAPEX) and operating expenditure (OPEX), for example through deployment of Circulating Fluidized Bed (CFB) boilers. This fuel quality penalty is modelled, relative to a wood-chip fired bioenergy plant, as an increase in base CAPEX for the combustion technologies (BES and BECCS) of 25% for high-ash feedstocks (ash above 5%) or 10% for medium-ash feedstocks (2–5%), and a reduction in electrical conversion efficiency of 10% or 5% respectively (23). O&M costs are not penalised separately: they are charged as a fraction of CAPEX, which already carries the penalty, and straw-fired plants have the same ratio of O&M to investment cost as wood-fired plants (24). The fouling, slagging and corrosion risks of high-ash, high-alkali residues are reviewed by (25), (26) and (27).

4.2.3 Regional pricing structures

Feedstock is priced at the field side (farm gate or road side), in 2024 USD per Mg; haulage from the field to the plant is costed separately (see Biomass transport costs and emissions), so delivered or plant-gate prices are not used. For Europe, China and India the values are costs of collection at the field side, excluding the margins of farmers and intermediaries; for the United States, where no comparable national cost estimate is available, the value is a farm-gate price that includes the grower’s payment and profit, so it is somewhat higher than a cost-only estimate would be.

- **United States:** US\$50 per dry short ton (2011 USD), or 76.9 US\$/Mg in 2024 USD: in the Billion-Ton Update, about 80% of the primary crop residue supply is profitable to collect at a farm-gate price of US\$45 per dry ton and 90% at US\$50 (28). The Billion-Ton prices are uniform across the country because the underlying POLYSYS runs apply a single national price, raised until a large and reliable supply is offered. The farm-gate price comprises the harvest cost and a grower payment that already compensates for the nutrients removed with the residue, so no separate nutrient-replacement charge is added.
- **Europe:** road-side costs of cereal straw from the S2Biom atlas (29), which maps them at NUTS-3 level by activity-based costing, aggregated to country level in three cost classes: €20 per t of dry matter (2012 EUR) for eastern Europe, Poland and the Baltic states; €37.5 for most of central, northern and southern Europe, including France and Spain; and €62.5 for Germany, Belgium, the Netherlands and Luxembourg. In 2024 USD these are 35.1, 65.8 and 109.7 US\$/Mg. Countries without an S2Biom value are assigned by analogy: Kosovo to the lowest class, like Serbia and Croatia; Cyprus to the middle class, like Greece; and Türkiye and the North African and Middle Eastern countries within the European grid to the lowest class, as residue costs there are generally below those of most of Europe.
- **China:** the field-side cost of collecting corn straw estimated with the StrawFeed supply-chain model for Nongan county, Jilin (30): raking, baling and loading cost ¥99.6 per t (2019), or 17.7 US\$/Mg in 2024 USD, excluding the ¥72.5 per t of transport to the plant. Like the European and Indian values, this is a cost and excludes the margins of farmers and brokers; power plants in Northeast China pay about ¥300 per t delivered (30), and ¥200–300 per t at the plant gate in other major producing provinces.
- **India:** paddy straw, which has little value as fodder in the rice–wheat belt of the north-west because of its high silica content and is otherwise burned in the field, is priced at 34 US\$/Mg on a dry basis, the cost of baled straw at the field side estimated for bioenergy supply chains in Haryana (31). The Monte Carlo analysis samples it between about 12 US\$/Mg, reflecting the low value of paddy straw where it is burned (32), and about 50 US\$/Mg, the price of straw traded as fodder (33, 34). Wheat straw, which is valued as fodder, is not used as a feedstock.

Year-round operation also requires interim storage of residues harvested within a few weeks. Published storage costs vary widely, from about 3 to 42 US\$/Mg (2024 USD): 3–11 for US farm-scale storage (14, 35, 36), 18 for on-farm and 31 for purpose-built storage in the United Kingdom (37), 34 for storage and handling (38), and about 42 for six months’ storage of German straw, where storage is 36.4% of the total

supply cost (39). We add 15 US\$/Mg for the United States and Europe, sampled from 3 to 42 in the Monte Carlo analysis, and scale it by the regional haulage cost factor for China and India (11 and 10 US\$/Mg), consistent with storage accounting for 20–30% of the delivered cost of paddy straw in India (40).

4.3 Technology pathways and thermodynamics

The core thermodynamic engine evaluates three distinct biomass utilization pathways, processing a biomass input of 125 MW_{th} by default (Table 1) into varying ratios of exported electrical energy and sequestered carbon.

4.3.1 Bioenergy (BES)

The BES pathway represents a traditional, dedicated biomass combustion facility. In this baseline scenario, the feedstock is fully oxidized to maximize electricity exported to the regional grid. Because there is no active carbon capture, the physical carbon embedded in the biomass is returned to the atmosphere.

The technoeconomic parameters for this modern combustion archetype assume a net electrical efficiency of 30% (23). Capital expenditures (CAPEX) are specified regionally per kW of net electrical capacity for a 50 MW_e wood-fired reference plant (Table 1), and scale non-linearly with plant capacity via a standard 0.7 scaling factor. The reference CAPEX (2024 USD) is 4,700 \$/kW_e for the United States, from the EIA’s 50 MW biomass plant (41), and 4,300 \$/kW_e for Europe, from Danish technology data for a 132 MW_{th} straw-fired plant scaled to 50 MW_e and adjusted to a wood-fired equivalent by the Danish straw-to-wood cost ratio (24). For China and India, where far more biomass plants have been built recently, it is the average installed cost of bioenergy plants commissioned in 2024, 2,200 and 1,900 \$/kW_e (42); India’s regulatory benchmark for rice-straw plants is lower still, about 925 \$/kW_e (43). For costing, plant capacity is defined as the thermal input multiplied by a fixed reference net efficiency (`bes_capex_ref_eff`), rather than by the realised efficiency. Equipment cost is therefore set by the size of the boiler and steam cycle, so factors that reduce net output for a given thermal input (the high-ash fuel penalty, efficiency uncertainty, or the parasitic load of carbon capture) reduce revenue without spuriously reducing capital cost. Operations and maintenance (O&M) costs, fixed and variable together, are charged as an annual fraction of total CAPEX (`plant_om_factor`, 4% per year), which is shared by all conversion technologies (BES, BECCS, and the pyrolysis unit and energy block of BEBCS) and the same in every region (see Regional cost adjustment). The 4% value is consistent with the EIA reference 50 MW biomass plant (fixed O&M of \$141.5/kW-yr plus variable O&M of \$5.44/MWh, about 3.5% of CAPEX per year at a 70% capacity factor) (44), with IRENA’s range for stoker and fluidized-bed biomass boilers (fixed O&M of 3.2–4.2% of CAPEX per year

289 plus \$3.8–4.7/MWh) (45), and with Danish technology data for 80 MW_{th} biomass CHP plants (4.4% per
 290 year for straw and 4.6% for wood pellets, fixed and variable combined) (24). Capital costs are annualized
 291 over the plant lifetime at the regional discount rate using the capital recovery factor, so that all costs and
 292 revenues are expressed as levelized annual values per megagram of feedstock; because O&M is a constant
 293 annual cost, discounting it over the plant lifetime and re-annualizing at the same rate returns the annual
 294 value. Total electrical output is derived directly from the biomass lower heating value (LHV, 18.6 GJ/Mg for
 295 dry, ash-free feedstock (46)) and the plant’s capacity factor. Technology data give an availability of 91% for
 296 biomass plants (23), but this accounts only for planned and unplanned outages, not for operation below full
 297 output, which depends on how each technology is dispatched. Capacity factors are therefore specified per
 298 technology (ranges sampled in the Monte Carlo analysis). Historical capacity factors of biomass power plants,
 299 such as 0.54–0.61 for US wood-fired plants over the past decade (47) and 0.50–0.64 for Drax’s biomass units
 300 in 2023–2024 (48), reflect dispatch in grids with low or no carbon prices and are a poor guide to operation in
 301 decarbonising grids. Models that optimise capacity and hourly dispatch together under carbon constraints
 302 find two roles for biomass power: firm, base-load low-carbon supply (49, 50), favoured at high CO₂ charges
 303 where other sources of flexibility are ample (8), and flexible supply in hours of high residual load in systems
 304 dominated by wind and solar (6, 7). BES is therefore evaluated in both modes in the same run, and each
 305 grid cell takes the mode with the higher net value: base load at a capacity factor of 0.85 (range 0.75–0.90)
 306 and flexible operation at 0.45 (range 0.30–0.60). Because a plant that follows prices sells in high-price hours,
 307 its realised price exceeds the time-averaged wholesale price. We estimated this ratio from hourly day-ahead
 308 prices in 24–31 European markets over 2015–2025 (51), for a plant that runs the highest-priced hours of each
 309 day among its available hours (availability 0.91), with perfect foresight and no start-up or minimum-load
 310 constraints. The median ratio across markets in 2024–2025 was 1.05 at a capacity factor of 0.85 and 1.30 at
 311 0.45 (10th–90th percentile 1.19–1.42); the latter has risen from 1.15–1.21 in 2015–2023 as solar generation
 312 has grown. The daily price spread available to a flexible generator is bounded by grid storage: prices in
 313 discharge hours cannot persistently exceed the charging price divided by the round-trip efficiency plus the
 314 levelised cost of storage, which at 65 US\$/MWh for utility-scale batteries (52) gives about 1.34 times the
 315 average European price (and about 1.9 at the US costs of (53)). Batteries do not shift energy between seasons
 316 or across multi-day lulls in wind, whereas stored biomass can, and models of decarbonised systems show a
 317 wider spread of prices with more hours at both very low and high prices (54). The ratios are therefore 1.05
 318 (range 1.02–1.08) for base load and 1.30 (range 1.15–1.45) for flexible operation, applied in all regions; they
 319 are derived from European markets and are an assumption elsewhere. Abatement per megagram of feedstock
 320 is the same in both modes, so the choice does not depend on the carbon price. The allocation between modes
 321 also depends on the generation mix and demand of each grid, which are not modelled. BECCS is driven

towards base-load operation by its higher capital cost, by removal revenue earned per tonne of CO₂ captured and by the need to keep the capture plant, dense-phase pipeline and injection in steady operation (6, 50); it takes 0.85, the design capacity factor of CCS power plants (55), with a range of 0.60–0.90 whose lower end reflects operation of about 3,900 full-load hours where scarce biomass is directed to high-price hours (7). Electricity is a small share of BEBCS revenue, so its operation is limited by the availability of the pyrolysis unit and the continuity of biomass supply rather than by dispatch; it takes 0.80 (range 0.70–0.85), below the 90% on-stream factor of nth-plant pyrolysis designs (56) to allow for interruptions of feedstock supply.

When ash recycling is enabled (the default; `ash_recycling`, Table 1), the bottom and fly ash from BES and BECCS combustion is returned to cropland and credited with its liming value, where soil pH is below the target pH, and its phosphorus fertilizer value, using the same substitution prices and availability factor as biochar (see Biochar agronomic valuation). Phosphorus is largely retained in the ash, whereas nitrogen and most of the potassium are volatilized during combustion and are not credited. The default ash liming (calcium carbonate equivalent) and phosphorus contents are set so that, at the default feedstock ash content, the ash and the biochar derived from the same feedstock return equal liming and phosphorus value per megagram of feedstock, so that the pathways differ only in the nitrogen, potassium and soil physical benefits retained by biochar.

4.3.2 Bioenergy with carbon capture and storage (BECCS)

The BECCS pathway augments the standard combustion baseline with a post-combustion carbon capture unit. The addition of this unit imposes a significant parasitic energy penalty, primarily driven by solvent regeneration and CO₂ compression requirements. Consequently, the net electrical conversion efficiency drops below the BES baseline by an efficiency penalty of 8 percentage points (range 5–13), at a capture rate of 90% (range 85–95%) (57). Consistent with costing BECCS as a mature technology, the central value reflects recent post-combustion amine capture: reviews place the penalty for pulverised-coal plants at 7.7–11.9 percentage points, and the Boundary Dam 3 unit lost about 8 points with capture (58); present technology loses about 10.8 points, falling to 7.2 with improved solvents, compression and heat integration (59). The lower bound reflects projected future penalties of 5.25–6.25 points (60) and the upper bound the 12.8–15.9 points estimated with 2009 amine technology for dedicated biomass plants (57).

The capture process is parameterized with a 90% capture efficiency rate. The total volume of captured CO₂ routed to geological storage is calculated stoichiometrically based on the carbon fraction of the incoming biomass feed (default 48% C (46)), converted to CO₂ mass equivalents using the 44/12 molar ratio. The additional infrastructure (capture unit, CO₂ compression and flue-gas cleaning) is represented as a premium

on total CAPEX relative to a BES plant of the same thermal input (`beccs_capex_premium`, Table 1). The default premium is based on the IEAGHG techno-economic evaluation of dedicated biomass-fired power plants with post-combustion amine capture, in which capture raised total investment by 38–40% for 75 MW_e bubbling and 250 MW_e circulating fluidized bed plants (57). Because net output fell simultaneously, the corresponding increase in specific capital cost was 114–126% per net kW_e, consistent with the 45–130% range reported across biomass CCS technologies by (61). Expressing the premium on a thermal-input basis keeps it independent of the efficiency penalty. Annual O&M is the same fraction of CAPEX as for the other technologies (Table 1).

4.4 Carbon transport and geological storage

4.4.1 Storage sites and sink choice

Geological storage is represented by 45 storage basins across the four regions (onshore saline formations, onshore oil fields suitable for CO₂-enhanced oil recovery (EOR), and offshore saline formations and depleted fields). Saline storage and EOR are flagged separately, so that petroleum basins with substantial saline storage resource (for example the Ordos, Songliao, Tarim and Junggar basins in China, the Cambay Basin in India and the Permian Basin in the United States) serve as saline sinks as well as EOR sites. They are listed with their locations and types in the model repository. Basin locations and storage types follow the OGCI CO₂ Storage Resource Catalogue (62) and, for the United States, the USGS national assessment of geologic CO₂ storage resources (63); basins added for coverage are located at operating or planned storage projects (for example Porthos, Northern Lights, Greensand, Ravenna CCS, HyNet and Bayou Bend) listed in the Global CCS Institute’s CO2RE facilities database (64). Four transport-and-storage classes are evaluated for every grid cell: pipeline to an onshore saline site; pipeline to an EOR site (only when EOR is allowed, which is not the default); pipeline and ship to an offshore site; and pipeline, including a subsea section, to an offshore site. For each cell and plant size, the model selects the class with the lowest combined transport and storage cost for that plant’s CO₂ flow.

4.4.2 Pipeline routing

Pipeline routes are computed on a grid ten times finer than the model grid (0.02° for the United States, 0.01° elsewhere). Two surfaces are used, one to choose each route and one to cost it, so that the routing can avoid difficult terrain without that avoidance being charged as extra pipeline length. Slope from the 250 m Geomorpho90m dataset (65) is converted, per 250 m pixel, into a construction-cost multiplier $F_{cost}(\theta)$. This multiplier is piecewise linear in slope θ : 1.0 at 0°, 1.05 at 5°, 1.2 at 10°, 1.6 at 17°, 2.3 at 25°, 3.2

at 35° and 4.0 at 45°. The per-pixel values are then averaged to the routing grid, with additive premiums above 2,500 m altitude. The routing surface adds a planning-risk premium for steep ground, $F_{route} = F_{cost} [1 + r_{max} \text{logistic}((\theta - 20^\circ)/3^\circ)]$ with $r_{max} = 2$, capped at 25.

Protected areas from the World Database on Protected Areas (66) are tiered. Strictly protected areas (IUCN categories Ia, Ib and II, and natural World Heritage sites) are barriers. Other designated terrestrial protected areas can be crossed at a construction-cost premium of 25% and a routing premium of 100%. Marine protected areas are not applied to the land routing.

Least-cost paths on F_{route} from every cell to the nearest target were computed with terra (12). The resulting route tree was then traced to measure, for each route, its physical length L , the line integral of F_{cost} along the route, and the highest point of the route above the source. Pipeline cost is charged on the physical length multiplied by the route-average construction-cost multiplier (the line integral of F_{cost} divided by L). No further tortuosity factor is applied. In a robustness test, setting r_{max} between 0 and 2 changed the optimal technology in less than 0.4% of grid cells.

4.4.3 Hub-and-spoke pipeline modality

Standard length-invariant TEA cost functions for CO₂ pipelines adequately model regional networks but underestimate the capital and energy burdens of long-distance transport, which requires intermediate booster compression to overcome frictional pressure drops. Pipeline CAPEX is therefore calculated with a piecewise-linear hub-and-spoke model:

1. **The Feeder Leg:** Sources build a dedicated feeder pipeline for the first 50 km (d_{feeder}), bearing the full CAPEX scaled to their own CO₂ flow (scaling exponent 0.6). Routes of 50 km or less are dedicated pipelines throughout.
2. **The Trunk Leg (Base Phase):** Up to 700 km, the source connects to a regional trunkline operating at an efficient scale (3 Mt CO₂ yr⁻¹), paying a share of its CAPEX proportional to its mass flow.
3. **The Trunk Leg (Booster Phase):** Beyond 700 km, the marginal trunkline cost is doubled ($P_{booster} = 2.0$) to approximate the capital and operating costs of intermediate booster stations.

The trunkline CAPEX share is:

$$CAPEX_{trunk} = m \cdot \begin{cases} C_{base} \cdot (L - 50) & \text{if } L \leq 700 \\ C_{base} \cdot (700 - 50) + (C_{base} \cdot P_{booster}) \cdot (L - 700) & \text{if } L > 700 \end{cases}$$

Here L is the physical route length (km), m is the route-average construction-cost multiplier (applied to the feeder leg as well), and C_{base} is the unit-length cost of the efficient-scale trunkline, scaled to the source's share of its mass flow. The reference cost is US\$50 million per 100 km for a 1 Mt yr⁻¹ pipeline, scaled by the regional CAPEX factor. This lies within the range given by the diameter-based cost function of (67) (2.16 € per m per mm of diameter, 2024 euros) for the 200–250 mm pipeline needed to carry 1 Mt yr⁻¹ of dense-phase CO₂ at typical velocities, 0.43–0.54 million € (0.47–0.58 million US\$) per km. Annual pipeline O&M is 4% of CAPEX, and pipeline CAPEX is annualized over the 25-year plant lifetime at the regional discount rate. Routes that climb require booster pumping to restore the pressure lost in lifting dense-phase CO₂. Pumps are sized for the lift beyond a 1 MPa design margin, with pump CAPEX following (68), and their electricity is charged at the regional electricity price.

4.4.4 Maritime shipping and subsea pipelines

CO₂ destined for offshore sinks is either shipped or carried by pipeline to the sink:

$$C_{ship} = C_{pipeline}(L_{port}) + C_{liquefaction} + C_{terminal} + c_{voyage} \cdot d_{sea}, \quad C_{subsea} = C_{pipeline}(L_{landfall} + d_{sea})$$

L_{port} and $L_{landfall}$ are the physical lengths of the onshore routes to the port or landfall. d_{sea} is the sea distance from there to the sink, routed on a 0.02–0.04° grid with land impassable. Liquefaction costs 24.5 US\$ per t CO₂, from the 21.1–21.3 US\$ per t (2021) of the least-cost liquefaction processes for ship transport at 15 bar (69), and the fixed cost of ships, buffer storage and terminals is 15 US\$ per t (both scaled by the regional CAPEX factor); the voyage costs 0.010 US\$ per t per km. The fixed and voyage costs are the intercept and slope of the ship transport cost (excluding liquefaction) of a 2.5 Mt yr⁻¹ point-to-point chain, 8.2 € per t at 180 km rising to 14.5 € per t at 1,500 km in 2009 euros (70), converted to 2024 US dollars. For subsea pipelines, the subsea section carries 1.5 times the onshore CAPEX per km, within the 1.4–1.7 range for offshore relative to onshore pipelines (71).

Every coastal cell is a candidate port or landfall. The land route and the port are chosen together, by a multi-source least-cost search in which each coastal cell starts with its sea leg already priced: 0.2 flat-onshore-pipeline-km per sea km for shipping (the ratio of voyage to trunk-pipeline cost per t km), and 1.5 for subsea pipelines. The route therefore minimizes land and sea cost together rather than land distance alone. Fixed liquefaction and terminal costs do not affect this choice.

Emissions from CO₂ transport are deducted from the BECCS abatement. For shipping, these are the

emissions from liquefaction, loading and the voyage, about 2.5% of the CO₂ carried over 200 km and about 18% over 12,000 km (72). For pipelines, they are the emissions from the booster pumping electricity, at the displaced grid intensity.

4.4.5 CO₂ injection

CO₂ injection and storage (injection wells and operations, and measurement, monitoring and verification) is costed separately from both the capture plant and CO₂ transport, as a fixed cost per megagram of CO₂ injected, with separate values for onshore and offshore storage sites (`ccs_storage_cost` and `cost_offshore_storage`, Table 1). The defaults, 10 US\$/t CO₂ onshore (range 2–33) and 20 US\$/t CO₂ offshore (range 5–50), follow a recent lifecycle assessment of storage costs from exploration to 50 years of post-closure monitoring. That assessment gives 2–15 US\$/t onshore and 5–31 US\$/t offshore (2024 USD) for reservoirs with open boundary conditions, rising to 5–33 and 8–50 US\$/t for moderate-quality closed reservoirs, and up to 58 and 147 US\$/t for the least favourable closed reservoirs (73). These ranges are consistent with earlier estimates for European saline aquifers of €2–12/t onshore and €6–20/t offshore (74). Because boundary conditions, which depend on the geology, are the main driver of storage cost (73), storage sites are assigned a site-specific cost where the evidence allows. For onshore saline formations in the United States, the cost is the first-year break-even price of the cheapest formation in the sink’s state with at least 1 Gt of prospective storage resource (in the basin, where no formation in the state qualifies), from a formation-by-formation model that accounts for pressure build-up and interference (75), escalated from 2023 to 2024 USD. These range from 6.4 US\$/t (Frio Formation, Gulf Coast) to 22.5 and 24.7 US\$/t in the Denver-Julesburg and Appalachian basins. Elsewhere, saline formations are classed as open where they are regionally extensive and well connected (the Utsira and Johansen formations of the northern North Sea, the Dogger aquifer of the Paris Basin and the Miocene sands of the Gulf of Mexico shelf), and as closed where they are compartmentalised (the Buntsandstein of the North German Basin), and take the midpoint of the corresponding range above (open: 8.5 US\$/t onshore and 18 offshore; closed: 19 onshore). Depleted oil and gas fields are not closed aquifers in this sense, because they start below their original pressure; the six depleted fields among the sinks take 16.2 US\$/t, the mean of ZEP’s offshore cases with and without re-usable wells (€6 and €10/t in 2009) (74) escalated to 2024 USD. Sinks without sufficient evidence, including all those in China and India, take the default onshore or offshore values. Site-specific costs are scaled by a common factor sampled from 0.5 to 1.75 in the Monte Carlo analysis (`storage_cost_factor`). The applicable cost follows the sink reached by the route chosen for each grid cell (see Storage sites and sink choice). Because storage costs differ between sinks, the route search starts each sink at its storage cost, expressed as an equivalent length of flat onshore trunk pipeline at 0.05 US\$/t/km (for example, 160 km for

8 US\$/t), so that a farther sink with cheaper storage can be chosen. Injection costs are applied to the mass of CO₂ captured, in addition to the transport cost and the capture plant’s own CAPEX and O&M.

4.5 Bioenergy with biochar storage (BEBCS)

Unlike the fully oxidative pathways, the BEBCS pathway relies on thermochemical pyrolysis, producing a solid, recalcitrant carbon matrix (biochar) alongside volatile syngas and bio-oil. The model employs a mass and energy balance approach constrained by elemental conservation of Carbon, Hydrogen, and Oxygen (C, H, O).

The mass yield and elemental composition of the resulting biochar are determined dynamically as a function of the peak pyrolysis temperature (T_{py}) and the lignin fraction of the feedstock. Specific yields for non-condensable gases hydrogen (H_2), carbon monoxide (CO), methane (CH_4), and ethylene (C_2H_4) are calculated based on temperature-dependent reaction kinetics (76). To close the mass balance, the unaccounted elemental fractions (U_c, U_h, U_o) are stoichiometrically partitioned into bio-oil, water (H_2O), and carbon dioxide (CO_2). Bio-oil C and H mass fractions are set at 1.25 and 1.26 times those of the feedstock (e.g. 62.5% C and 7.56% H for a feedstock with 50% C and 6% H), allowing the final product distribution to be solved analytically (76). The default peak pyrolysis temperature of 500 °C is a modelling assumption representing the medium-temperature class (450–600 °C) of (77) and the IPCC inventory guidance (78); it is sampled $\pm 10\%$ in the Monte Carlo analysis. All yields and energy flows are expressed per megagram of dry, ash-free feedstock, with feedstock ash reporting to the biochar.

The net usable energy flux for the BEBCS pathway is calculated from an explicit energy balance on a consistent higher heating value (HHV) basis. The process heat demand is the HHV of the products (biochar, bio-oil and permanent gases) plus explicit process heat losses, minus the HHV of the feedstock. The HHVs of biochar and bio-oil are calculated via the Dulong formula from their C, H and O fractions, and that of the gas from the heats of combustion of its components. Heat losses comprise conduction through the insulated reactor wall (sized from the plant feed rate and a temperature-dependent solids residence time), the sensible heat of biochar discharged at the pyrolysis temperature, and the latent and sensible heat of the bio-oil vapour, permanent gases and water vapour (feedstock moisture plus reaction water) leaving the heat-recovery system. Because the water vapour term includes the latent heat of all water leaving the process, expressing both feedstock and products on an HHV basis avoids double-counting that latent heat. Process heat is supplied by combusting the syngas and bio-oil vapours in a process heater (80% efficiency, HHV basis), and the plant’s parasitic electricity demand (0.07 GJ_e per Mg feed) is met from the same fuel (76). If the syngas and bio-oil are insufficient to meet these loads, a portion of the biochar is oxidized to close the

energy balance, proportionally reducing the final carbon sequestration yield. The remaining fuel, converted to a lower heating value basis, is used to generate electricity in a dedicated power block (35% net electrical efficiency, LHV basis (76)), whose capital cost is sized on its actual fuel input using the same reference-plant scaling as the combustion pathways. A power block is not always worthwhile: where electricity prices are low or the displaced grid intensity is small, its capital cost can exceed its revenue and grid credit. BEBCS is therefore evaluated in each grid cell both with the power block and without any energy co-product, in which case the surplus vapours and gases are flared (process heat and the parasitic load are still met from the pyrolysis fuel), and takes the configuration with the higher net value at the prevailing carbon price.

The pyrolysis unit CAPEX is 335 US\$ per Mg yr⁻¹ of feed capacity in the United States (2024 USD), for the feed of a 50 MW_e reference plant, from NREL’s nth-plant estimate for a 2,000 dry t day⁻¹ slow pyrolysis plant (79) scaled to the reference size with an exponent of 0.7. The power block, which burns the pyrolysis vapours and gases and raises steam for a condensing turbine, costs 3,250 US\$/kW_e in the United States. This lies between the cost of a steam cycle alone in NREL’s pyrolysis designs, about 2,650 US\$/kW_e (56), and that of a complete 50 MW biomass power plant, which also includes solid-fuel handling (4,700 US\$/kW_e; (41)). Both costs are scaled to other regions by the capital cost location factor (see Regional cost adjustment).

4.5.1 Biochar permanence

The Bioenergy with Biochar Storage (BEBCS) pathway diverts a fraction of the original biomass carbon into a recalcitrant solid matrix. While physical disposal methods such as the landfilling of biochar represent a highly effective method to provide highly stable carbon storage, this framework strictly evaluates soil application to quantify the cascading agronomic co-benefits and regional substitution values. Consequently, the carbon retention within the soil matrix is finite and must be modeled dynamically based on localized biogeochemical conditions.

Biochar carbon permanence (F_{perm}). To accurately quantify the net carbon dioxide removal (CDR) of the BEBCS pathway, the model calculates the permanent carbon fraction (F_{perm}) that remains stabilized in the soil after a 100-year time horizon. This fraction is derived using the multi-pool decay methodology established by (77).

The baseline degradation rates of the biochar’s labile and recalcitrant carbon pools are fitted to either the peak pyrolysis temperature (T_{py}) or the molar Hydrogen-to-Organic-Carbon (H:C) ratio of the final char. The model uses the H:C ratio, derived from the temperature-dependent biochar composition, so that biochar yield and permanence respond jointly to T_{py} . Because microbial and chemical degradation kinetics are highly temperature-dependent, these reference decay rates are spatially adjusted to match the localized mean annual

soil temperature using a Q_{10} temperature sensitivity function. To optimize spatial computational efficiency across the high-resolution rasters, the thermodynamic module utilizes bilinear interpolation over a pre-calculated matrix of F_{perm} values bounded by soil temperatures (-55°C to 40°C) and H:C ratios (0.0 to 0.7).

Total carbon sequestered per megagram of feedstock is therefore modeled as:

$$C_{seq} = Yield_{bc} \cdot C_{content} \cdot F_{perm} \cdot \left(\frac{44}{12}\right)$$

where $Yield_{bc}$ is the mass fraction of biochar produced, $C_{content}$ is the carbon mass fraction of the biochar, and the 44/12 ratio converts the elemental carbon to CO_2 mass equivalents.

Soil N_2O emission reduction. Biochar additions to soil can suppress nitrous oxide (N_2O) emissions from nitrogen fertilizer. From field trials with biochar additions exceeding 10 Mg C ha^{-1} , (77) estimated a 23% reduction in N_2O emissions (95% confidence interval 5–41%). The model credits this fractional reduction (R_{N_2O}) against direct fertilizer-induced N_2O emissions on land receiving biochar at that application rate (A_{bc}), where direct emissions are the regional mineral N application rate (N_{app}) multiplied by the IPCC Tier 1 emission factor (EF_1) (80). Because significant effects have only been demonstrated in the first year after application (77), the reduction is assumed to persist for a short period (t_{N_2O}), with a long upper tail in the Monte Carlo analysis to reflect the uncertainty. The credit per megagram of feedstock is:

$$A_{N_2O} = \frac{N_{app} \cdot EF_1 \cdot R_{N_2O} \cdot t_{N_2O} \cdot C_{bc}}{A_{bc}} \cdot \frac{44}{28} \cdot GWP_{N_2O} \cdot 10^{-3}$$

where C_{bc} is the biochar carbon yield (Mg C per Mg feedstock), GWP_{N_2O} is the 100-year global warming potential of N_2O (81), and parameter values are given in Table 1. Because the effect is short-lived and spread over the large area to which biochar is applied, this credit is small relative to the carbon stored in the biochar.

4.5.2 Biochar agronomic valuation

Previous spatial models frequently value biochar using a static, exogenous global market price. To better reflect real-world agricultural economics, C-SCAPE endogenously determines the economic credit of biochar through a mechanistic substitution valuation. This assumes that the shadow price of biochar is equal to the cost of the conventional agricultural inputs it offsets:

$$Value_{ag} = Value_{lime} + Value_{nutrients} + Value_{physical}$$

1. **Liming Equivalence** ($Value_{lime}$): The model spatially evaluates the native soil pH against a neutral agricultural target (pH 6.5). If the soil is acidic, the biochar's Calcium Carbonate Equivalent (CCE, default 15%) is credited against the cost of the agricultural lime it displaces. Because substitution also avoids hauling and spreading the lime, this cost includes transport and application. It is the median of published values, each converted to a calcium carbonate equivalent of 100% and escalated to 2024 US dollars: 59 \$/Mg, ranging from 31 \$/Mg for lime delivered to Brazilian pastures, excluding spreading (82), through 52 \$/Mg in Oklahoma (83) and 59–110 \$/Mg near to and far from a quarry in Oklahoma (84), to 85 \$/Mg spread in North Carolina (85). The range is sampled in the Monte Carlo analysis. If the regional soil is already alkaline, the liming value is strictly zero.
2. **Nutrient Substitution** ($Value_{nutrients}$): The available nitrogen (N), phosphorus (P), and potassium (K) fractions within the biochar are credited against prevailing regional fertilizer costs. Nutrients are valued at the world market prices of standard commercial fertilizers: urea for N (1.06 US\$/kg N), triple superphosphate for P (1.27 US\$/kg P_2O_5) and potassium chloride for K (0.86 US\$/kg K_2O). These are averages of the World Bank's annual prices for 2021–2025, each escalated to 2024 USD (86). The same prices are used in every region: retail subsidies (as in India) and export controls (as in China) move farm-gate prices away from world prices, but they transfer value rather than change the economic value of the fertilizer displaced. Elemental P and K contents are converted to their oxide equivalents ($\times 2.29$ for P_2O_5 and $\times 1.20$ for K_2O) before pricing, and each nutrient is credited at its first-year plant availability (N 10%, P 50%, K 80%; Table 1).
3. **Physical/CEC Improvements** ($Value_{physical}$): Biochar raises the cation exchange capacity (CEC) of soils, improving their retention and supply of nutrients and water; the effect is largest in sandy or degraded soils. Unlike liming, this benefit is valued as a yield increment rather than as a substitute input. Agricultural lime is an equivalent substitute for biochar (or ash) as a pH amendment, so the liming benefit is credited at the cost of the lime it displaces. Fertilizer, however, is not a substitute for CEC but a complementary input: a soil with low CEC cannot retain and supply nutrients effectively and responds poorly to fertilizer, so raising its CEC lifts the whole fertilizer response curve, including the economically optimal application rate. The annual yield benefit is 21.1 US\$ per Mg of biochar carbon in soils with a CEC of 5 cmol/kg or less. It is derived from the mean crop-yield benefit of 30 US\$/Mg C per year in (18) (average 2008–2013 prices), of which about half is attributable to CEC and half to pH (the latter credited separately as the liming value), escalated to 2024 USD. The benefit

falls linearly with increasing CEC to zero at 30 cmol/kg; this linear scaling is a heuristic.

Liming and nutrient values are one-off credits per megagram of biochar applied. The CEC of biochar increases rather than declines as it ages in soil, through oxidation and functionalization of its surfaces, so the physical benefit is treated as persisting for as long as the biochar carbon remains. Its present value is the annual benefit divided by the sum of the regional discount rate and the decay rate of biochar carbon, the latter being the mean rate implied by the 100-year permanence factor ($k = -\ln F_{perm}/100$; see Biochar permanence).

4.6 Baseline energy system and offsets

To robustly account for the climate mitigation benefit of bioenergy generation, the framework evaluates greenhouse gas offsets using a consequential Life Cycle Assessment (LCA) boundary. Total carbon abatement is calculated distinctively for each technology pathway, balancing active sequestration, grid displacement credits, and Scope 3 supply chain emissions.

4.6.1 Net carbon abatement balances

The net carbon abatement for each grid cell is defined as the sum of physical carbon sequestered and emissions displaced, minus the transportation emissions penalty:

- **BES:** Abatement relies entirely on the displacement of grid emissions, minus the Scope 3 emissions associated with diesel trucking of the biomass feedstock.
- **BECCS:** Abatement includes the physical CO₂ captured and injected into the geological sink (assuming a 90% capture rate) plus the grid displacement credit. However, because the parasitic energy load of the capture process reduces the net electrical efficiency, the total energy exported and thus the resulting grid displacement credit is proportionally lower for BECCS than for pure BES.
- **BEBES:** Abatement combines the grid displacement credit with the permanent CO₂ equivalent of the biochar carbon retained in the soil (F_{perm}) and the avoided soil N₂O emissions (A_{N_2O}), minus the diesel emissions of biochar field application.

For BECCS, the emissions of CO₂ transport are also deducted (see Maritime shipping and subsea pipelines). The residue removal counterfactual below applies to all pathways.

For all pathways, Scope 3 feedstock transport emissions are modeled dynamically by applying a heavy-duty diesel transport emissions factor (0.0001 Mg CO₂e per Mg · km, the well-to-wheel default for a 34–40 t articulated diesel truck at average load, including empty running (87)) to the effective biomass collection

distance.

4.6.2 Residue removal counterfactual

Removing crop residues for energy has two further effects on greenhouse gas balances that are common to all pathways, relative to a counterfactual in which residues remain in the field. First, retained residues add to soil organic carbon (SOC); residue retention increases SOC stocks by around 10% relative to removal (88). Retained residues also add nitrogen to the soil and increase soil N₂O emissions. Process-based simulations across more than 400 million hectares of cropland found that, through to 2100, the SOC gain and the N₂O increase from retaining more residue approximately cancel almost everywhere, the Brazilian Cerrado being the main exception (9). The net soil GHG penalty of removal (P_{soil} , Mg CO₂e per Mg feed) is therefore set to zero by default, and retained as a parameter for sensitivity analysis (Table 1). A penalty that counted the forgone SOC alone, without the offsetting N₂O, would be about 0.19 Mg CO₂e per Mg feed. Second, in regions where residues are commonly burned in the field, removal avoids the non-CO₂ emissions from open burning (CO₂ from burning is biogenic and is not counted). These are estimated with the IPCC Tier 1 emission factors for methane and nitrous oxide from burning agricultural residues and the associated combustion factor (89), applied to the share of the available residue in each grid cell that would otherwise be burned (f_{burn}). The burnt mass is estimated from the burnt share of cereal residue production in each 0.5° cell (2017–2021 mean) of a global cereal residue dataset that combines satellite fire observations with residue production (90), multiplied by the cereal field residue production underlying the residue supply (10); f_{burn} is this mass divided by the available residue, both averaged to the model grid, and capped at 1. Using the burnt share rather than the burnt mass keeps the different residue totals of the two datasets out of the ratio. Burning of non-cereal residues, such as sugarcane trash, is not counted, and satellite fire products under-detect small field fires, so the estimate is conservative; a multiplier sampled from 0.5 to 1.5 represents this uncertainty in the Monte Carlo analysis. The biomass-weighted mean of f_{burn} is 2% in the United States, 6% in Europe, 3% in China and 14% in India, and burning is concentrated in Punjab and Haryana in India, Ukraine and the Balkans in Europe, and north-eastern China:

$$A_{residue} = f_{burn} \cdot CF \cdot (EF_{CH_4} \cdot GWP_{CH_4} + EF_{N_2O} \cdot GWP_{N_2O}) \cdot 10^{-3} - P_{soil}$$

where CF is the combustion factor, EF the emission factors (g kg⁻¹ dry matter), and GWP the 100-year global warming potentials (81); parameter values are given in Table 1. Because the same term applies to every pathway, it does not change which technology is preferred in a grid cell, but it changes the absolute

net abatement and therefore shifts the marginal abatement cost curves.

4.6.3 Grid displacement

The carbon intensity of the electricity displaced by bioenergy output depends on how far the energy system has decarbonized, which in turn depends on the carbon price. Because the carbon price P is an exogenous input to the analysis, the displaced marginal emission factor (MEF) is modeled as a continuous function of P , so that the model does not combine a high carbon price with a fossil-dominated grid margin. Each grid cell’s empirical build margin at today’s explicit carbon price is used as an anchor and scaled by a regional price-response curve calibrated to integrated assessment scenarios.

Empirical anchor. New bioenergy capacity is assumed to displace the technology mix currently expanding on each grid. Using monthly national and US state electricity generation data (91), the sources with positive net generation growth ($\Delta G_i > 0$) between 2019 and 2024 are identified, and the build-margin intensity is their growth-weighted mean:

$$MEF_{emp} = \frac{\sum_i \Delta G_i \cdot CI_i}{\sum_i \Delta G_i}$$

where CI_i is the IPCC median life-cycle greenhouse gas intensity of source i (92): coal (820 g CO₂e/kWh), natural gas (490), solar (48), geothermal (38), hydro (24), nuclear (12) and wind (11.5). Bioenergy generation is assigned zero intensity, because bioenergy emissions are accounted for explicitly in the model; growth in bioenergy generation still counts in the denominator. Where no growth data are available, the average grid intensity is used. The resulting anchors are 378 g CO₂e/kWh for China and 574 for India, with medians over biomass-producing grid cells of 341 for the United States (state level) and 38 for Europe (country level).

Price-response curve. The decline of the displaced intensity with carbon price follows a Hill (log-logistic) curve,

$$MEF(P) = MEF_{min} + (MEF_{emp} - MEF_{min}) \frac{H(P)}{H(P_{now})}, \quad H(P) = \frac{1}{1 + (P/P_{50})^k}$$

where P_{now} is the region’s current explicit carbon price on power-sector emissions (Table 1: the EU Emissions Trading System and China’s national emissions trading scheme; no national price in the United States or India), so that the curve passes through each cell’s empirical anchor at today’s price. The scenario carbon price is thus interpreted as an explicit price: for a given explicit price, the curve gives the intensity of the electricity that new bioenergy capacity would displace, with today’s non-price policies embedded in the

empirical anchor. Prices below today’s level imply a more carbon-intensive margin than observed, capped at the intensity of coal. $MEF_{min} = 20$ g CO₂e/kWh is the residual life-cycle intensity of a decarbonized grid (93). The midpoint price P_{50} and steepness k were fitted for each region to the NGFS Phase 5 scenarios (seven scenarios; GCAM 6.0 and MESSAGEix-GLOBIOM 2.0) (94), by robust least squares with equal model weights. GCAM reports the United States, India and the European Union as native regions. For MESSAGEix-GLOBIOM, whose native regions are larger aggregates, the NGFS country-level downscaled results for the United States, India and the EU-27 were used, with coal and gas generation divided into unabated and CCS shares according to the parent region. For each model, the fleet-average intensity of electricity generation at each five-year step was normalized by that model’s own low-price value, so that only the shape of the decline, and not its level, is transferred to the empirical anchors. The fleet average was used rather than the models’ build margin because, in these scenarios, new capacity is already predominantly renewable at zero carbon price, leaving the build margin with little price response; the fleet average tracks the decarbonization of the system that a baseload plant displaces over its lifetime. A two-stage variant (coal-to-gas switching, then decarbonization) was selected by the Akaike information criterion for Europe only; single-stage curves were imposed for China and India, where coal-to-gas switching is limited by gas availability. The fitted midpoints are $P_{50} = 103$ US\$/t CO₂ for the United States, 53 for China and 44 for India, and 45 and 138 US\$/t for the two stages in Europe (2024 USD). REMIND-MAgPIE was excluded from the fit because its grid decarbonizes over time even at low carbon prices, so that its price response could not be separated from its time trend. Parameter uncertainty is represented by 1,000 bootstrap draws over models and scenarios, which are sampled in the Monte Carlo analysis. Because the same carbon price values removals and emissions, the displaced intensity declines as the removal price rises.

4.7 Regional cost adjustment

The combustion plant CAPEX of BES and BECCS is specified for each region directly (see Bioenergy (BES)). Other capital costs (pyrolysis plants, the BEBCS power block, pipelines, liquefaction and port terminals) are specified for a base location and scaled by a regional capital cost location factor, so that the relative cost of technologies is consistent within each region. O&M is charged at the same fraction of CAPEX in every region, so its absolute level follows regional CAPEX. The fraction does not fall where labour is cheaper: India’s regulated O&M allowance for biomass plants is 7.8% of the regulated CAPEX per year (43), compared with about 4% in the United States and Europe (24, 41). Biomass haulage costs are scaled by a regional road-freight cost factor (Table 1).

4.8 Spatial input data

Crop residue availability is the sustainably harvestable residue of (10) (net of residues retained for soil protection), minus the regional fraction used as livestock fodder and bedding (11). Feedstock composition is representative of cereal straws, corn stover and rice straw in the Phyllis2 database (46): 48% carbon and a lower heating value of 18.6 GJ/Mg on a dry, ash-free basis, with ash 5% of dry mass in the United States and Europe, where wheat straw and corn stover predominate, and 15% in China and India, where rice straw makes up a large share. Soil pH and cation exchange capacity (0–5 cm) are from SoilGrids 2.0 (95) and mean annual soil temperature (0–5 cm) from the global soil bioclimatic maps of (96). Electricity is valued at the wholesale price a generator receives, rather than at retail tariffs, whose regional differences largely reflect taxes, network charges and cross-subsidies. Prices are averages over 2024–2025, excluding the anomalous prices of the 2022 European gas crisis and its aftermath in 2023. For the United States, each state takes the price at the wholesale trading hub of the market that serves most of it (11 hubs) (97). For Europe, prices are national day-ahead market prices (98), with those of Türkiye (99) and Ukraine (100) from their market operators; small markets without published prices take a neighbouring market’s price. Belarus, which has no wholesale market, takes its industrial tariff (101) multiplied by the ratio of wholesale to industrial prices in the United States (median across states in 2024, 0.42) (97, 102). For China, the price is the provincial benchmark on-grid tariff for coal-fired power (103), which is also the reference price for renewable generators, excluding value-added tax. For India, it is the average price on the day-ahead market of the Indian Energy Exchange in 2024–25 (104), a single national value because the market rarely separates into regional prices. Each year’s price is converted at that year’s average exchange rate (105) and escalated to 2024 US dollars. The biomass-weighted mean prices are 40 \$/MWh for the United States, 96 for Europe, 47 for China and 49 for India. Grid carbon intensities are described under Grid Displacement.

4.9 Economic evaluation and technology selection

For each grid cell and technology, the net value per megagram of feedstock is the sum of revenues (electricity sales, the carbon price multiplied by the net abatement, and the agronomic value of biochar or ash) minus costs. The costs are annualized CAPEX, O&M, feedstock and haulage, and, for BECCS, CO₂ transport and storage. CAPEX is annualized with the capital recovery factor at the regional discount rate over the technology lifetime (25 years for BES and BECCS (23); 20 years for the pyrolysis unit and 25 years for its power block (18)). The regional discount rates are real, after-tax costs of capital for bioenergy projects: 5.6% for the United States, 6.8% for Europe, 6.2% for China and 7.1% for India. They are the nominal costs of capital in US dollars projected for 2030 by (106), converted to real terms with an expected inflation rate of

2.5%. We use after-tax rates, as is usual in techno-economic assessment. Grossing them up to pre-tax rates at statutory corporate tax rates (107) would add 3–4 percentage points, but would overstate the effective tax burden, which depreciation allowances reduce. Real rates are required because all cash flows, including the carbon price, are expressed in constant 2024 US dollars. All costs are in 2024 US dollars: values from other years are escalated with the US consumer price index (CPI-U), and values in other currencies are converted at the average exchange rate of the source year before escalation. The plant has a thermal input of 125 MW_{th} by default, and the preferred technology in each cell is the one with the highest net value. Because the displaced grid intensity falls as the carbon price rises (see Grid displacement), net abatement, and hence net value, is not linear in the carbon price. We therefore evaluate the model on a grid of carbon prices (0–250 US\$/t CO₂ in steps of 5 US\$/t, and 275–500 US\$/t in steps of 25 US\$/t) and interpolate abatement linearly between grid points; net value is the carbon-price-independent part plus the carbon price multiplied by the interpolated abatement (for BEBCS, both parts are those of the configuration chosen at each price). The break-even carbon price of a technology is the lowest carbon price at which its net value becomes non-negative. Marginal abatement cost curves assign each grid cell, at each carbon price, to the technology with the highest net value if that value is non-negative, and sum the biomass, area and abatement of the assigned cells.

4.10 Uncertainty and sensitivity analyses

Parameter uncertainty was propagated by Monte Carlo simulation, with 5,000 draws per region. Each parameter with an uncertainty range was sampled from a PERT or uniform distribution, with the central values, distributions and bounds listed in Table S1 (see Statistical methods), and correlated parameters were sampled jointly through a Gaussian copula. The uncertainty of the grid displacement curve was represented by sampling one of its 1,000 bootstrap draws in each iteration. The drivers of each technology’s net value were identified by fitting gradient-boosted regression trees [XGBoost; (108)] to the Monte Carlo results and computing SHAP values (109). The same method was applied across grid cells, to the choice of preferred technology, to identify its spatial drivers. A full factorial analysis evaluated 720 scenarios: four regions, carbon prices of 0–200 US\$/t CO₂, real discount rates of 2%, 5% and 10%, plant sizes of 50, 150 and 250 MW_{th}, with and without CO₂-EOR, and with and without early adoption. Under early adoption, each plant builds a dedicated pipeline over the full distance to the sink instead of sharing a trunkline.

4.11 Statistical methods

Table S1 lists every sampled parameter with its central value, distribution and bounds. The table is generated directly from the model’s canonical parameter file (`BiocharAG/inst/extdata/parameters.csv`), so it always matches the values used in the simulations. Ranges marked “ $\times$ central” are multipliers of the central value (which may differ by region); other ranges are in the parameter’s units. Where a bound differs by region it is shown in parentheses. For spatially explicit inputs (for example the electricity price), the relative range is applied as a multiplier on the spatial layer.

To identify which spatial inputs determine the optimal technology in each grid cell, we fitted a surrogate classifier to the spatial sensitivity results (scenario CP100_MW250: carbon price 100 US\$/t CO₂, plant size 250 MW_{th}) and attributed its predictions to the inputs with SHAP (SHapley Additive exPlanations) values (108). The classifier is a gradient-boosted tree ensemble (XGBoost; multi-class soft-probability objective, maximum depth 6, learning rate 0.05, 120 rounds) that predicts the technology with the highest net value (BES, BECCS or BEBCS) from ten cell-level inputs: biomass density, soil temperature, electricity price, field-side feedstock cost, the length of the CO₂ route chosen by BECCS, whether that route ends at an offshore sink, the storage cost at that sink, soil pH, soil cation exchange capacity (CEC) and the displaced grid carbon intensity. SHAP values are in log-odds of each class.

Two kinds of model are fitted. A global model, fitted to all four regions together, shows which inputs distinguish the regions as well as cells within them. Because some inputs are constant within a region (for example, feedstock cost outside Europe and the electricity price in India), in the global model they partly act as labels for the region rather than as drivers of within-region variation. A separate model is therefore fitted for each region; inputs that are constant within the region are dropped. The regional models are used for the maps of the dominant driver (Figure 7, Figure 8) and for the regional beeswarm and dependence plots (Figure S2 to Figure S6 and the dependence plots that follow them); Table S2 gives the share of each input in the mean absolute SHAP value of each model. Beeswarm and dependence plots show a random sample of 20,000 cells per model; SHAP values were computed for all cells. In the dependence plots, points are coloured by the input with the strongest estimated interaction with the plotted input.

4.12 Model parameters and data sources

To ensure full reproducibility and transparency, the operational, financial, and biophysical parameters utilized within the C-SCAPE framework are tabulated below. Parameters are classified by their spatial resolution: **Scalar** values are held constant globally; **Regional** values act as constants but are overridden based on the

785 macro-economic domain (e.g., India, China, Europe, US); and **Spatial** values are resolved dynamically at
786 the grid-cell level using high-resolution raster datasets.

Table 1: Operational, financial, and biophysical parameters of the C-SCAPE model.

Parameter	Description	Value				Units
		USA/Default	Europe	China	India	
Biochar & Agronomic						
avail_k	First-year potassium availability	0.8				fraction
avail_n	First-year nitrogen availability	0.1				fraction
avail_p	First-year phosphorus availability	0.5				fraction
bc_ag_value	Biochar agronomic value	0				\$/Mg
bc_cce	Calcium Carbonate Equivalent	0.15				fraction
bc_cec_value	Annual crop-yield benefit of biochar from raising soil CEC, at CEC <= 5 cmol/kg, falling linearly to 0 at CEC >= 30; valued as a perpetuity, annual value / (discount rate + biochar C decay rate)	21.1				\$/Mg biochar C/yr (2024 USD)
ash_cce	Calcium carbonate equivalent of recycled combustion ash	0.85				fraction
ash_p_content	Phosphorus content of recycled combustion ash (elemental P)	0.012				fraction
bc_k_content	Biochar potassium content (elemental K)	0.005				fraction
bc_n_content	Biochar Nitrogen content	0.005				fraction
bc_p_content	Biochar phosphorus content (elemental P)	0.002				fraction
bc_price	Biochar price	100				\$/t Biochar
bc_stab_factor	Biochar stability factor	4.6				factor
n_app_rate	Mineral N fertiliser application rate on cropland	60	74	198	118	kg N/ha/yr
n2o_ef	Direct N2O emission factor for fertiliser N	0.01				kg N2O-N / kg N
n2o_reduction	Fractional reduction in soil N2O emissions after biochar application	0.23				fraction
n2o_years	Duration of biochar N2O reduction	2				years

Table 1: Operational, financial, and biophysical parameters of the C-SCAPE model.

Parameter	Description	Value				Units
		USA/Default	Europe	China	India	
bc_app_rate_c	Biochar application rate for N2O effect	10				Mg C/ha
bc_field_cost	Biochar field application cost (spreading and incorporation)	116				\$/ha (2024 USD)
bc_field_diesel	Tractor diesel use for biochar spreading and incorporation	17.5				L/ha
price_k	Market price for potassium (world market; all regions)	0.86				\$/kg K2O (2024 USD)
price_lime	Cost of agricultural lime delivered and spread, per Mg of calcium carbonate equivalent (100% ECCE)	59				\$/Mg CaCO3-eq (2024 USD)
price_n	Market price for nitrogen (world market; all regions)	1.06				\$/kg N (2024 USD)
price_p	Market price for phosphorus (world market; all regions)	1.27				\$/kg P2O5 (2024 USD)
target_ph	Target agricultural soil pH	6.5				pH
time_frame	Time horizon of pyC permanence	100				years
CO~2~ Transport & Storage						
ccs_distance	CCS transport distance	100				km
co2_pipeline_terrain_share	Share of CO2 pipeline CAPEX that scales with the route-average terrain cost multiplier (v2 transport layers)	1				fraction
co2_subsea_capex_factor	Subsea CO2 pipeline CAPEX per km relative to onshore (offshore sinks reached by pipeline)	1.5				ratio
ccs_storage_cost	Onshore injection and monitoring (exploration to post-closure), for sinks without a site-specific cost	10				\$/Mg CO2 (2024 USD)
cost_offshore_storage	Offshore injection and monitoring (exploration to post-closure), for sinks without a site-specific cost	20				\$/Mg CO2 (2024 USD)
storage_cost_factor	Multiplier on site-specific CO2 storage costs (classified sinks in co2_sinks)	1				ratio

Table 1: Operational, financial, and biophysical parameters of the C-SCAPE model.

Parameter	Description	Value				Units
		USA/Default	Europe	China	India	
co2_ship_emis_fixed	CO2 emitted per tonne shipped, fixed part (liquefaction, loading, intermediate storage)	0.022				t CO2 / t CO2
co2_ship_emis_per_km	CO2 emitted per tonne shipped per km of voyage	1.3e-05				t CO2 / t CO2 / km
co2_liquefaction_cost	CO2 liquefaction for ship transport (scaled by capex_location_factor)	24.5				\$/t CO2 (2024 USD)
co2_ship_terminal_cost	Fixed ship transport cost per tonne: ships, buffer storage and terminals (scaled by capex_location_factor)	15				\$/t CO2 (2024 USD)
co2_ship_voyage_cost	CO2 ship voyage cost per tonne per km	0.01				\$/t/km (2024 USD)
Feedstock & Logistics						
bm_ash	Biomass ash mass fraction	0.05	0.05	0.15	0.15	fraction
bm_c	Biomass carbon mass fraction	0.48				fraction
residue_soil_ghg_penalty	Net soil GHG penalty of removing residues (forgone SOC minus avoided soil N2O)	0				Mg CO2e / Mg feed
residue_burn_fraction	Share of removed residues that would otherwise be burned in the field (regional fallback where residue_burn_map is absent)	0.02	0.06	0.03	0.14	fraction
residue_burn_factor	Multiplier on the cell-level burnt share (residue_burn_map)	1				ratio
residue_burn_cf	Combustion factor for field burning of residues	0.8				fraction
residue_burn_ch4_ef	CH4 emission factor for field burning of residues	2.7				g CH4 / kg dm
residue_burn_n2o_ef	N2O emission factor for field burning of residues	0.07				g N2O / kg dm
bm_h	Biomass hydrogen mass fraction (daf)	0.06				fraction
bm_h2o	Biomass moisture content	0.1				fraction

Table 1: Operational, financial, and biophysical parameters of the C-SCAPE model.

Parameter	Description	Value				Units
		USA/Default	Europe	China	India	
bm_lhv	Biomass lower heating value (dry ash-free)	18.6				GJ/Mg daf
bm_transport_fixed	Fixed loading/handling cost	6.27				\$/Mg (2024 USD)
bm_transport_var	Variable trucking cost per km of one-way haul (includes the empty return)	0.19				\$/Mg/km (2024 USD)
feedstock_storage_cost	Interim storage of seasonally harvested residues for year-round supply (about six months), US/EU basis; scaled by haulage_location_factor	15				\$/Mg (2024 USD)
india_feedstock_cost	Field-side price of baled paddy straw in India, dry basis (used for India only; other regions use calcu- late_regional_feedstock_cost() values)	34				\$/Mg (2024 USD)
haul_time_share	Share of variable trucking cost that is time-based (driver wages and benefits; truck payments; insurance; permits)	0.64				fraction
haul_fuel_share	Share of variable trucking cost that is fuel	0.24				fraction
lignin	Lignin fraction	0.2				fraction
transport_emissions_factor	Heavy-duty truck emissions	0.0001				Mg CO ₂ e / Mg.km
Financial & Market						
c_price	Carbon abatement credit price	50				\$/tCO ₂ e
capacity_factor	Reference capacity factor: biomass collection-distance layers are built at this value, and it applies where no technology-specific value is set	0.7				fraction
capacity_factor_bes_base	BES capacity factor in base-load operation	0.85				fraction
capacity_factor_bes_flex	BES capacity factor in flexible (load-following) operation	0.45				fraction

Table 1: Operational, financial, and biophysical parameters of the C-SCAPE model.

Parameter	Description	Value				Units
		USA/Default	Europe	China	India	
bes_base_price_capture	Ratio of realised to time-averaged wholesale electricity price for BES in base-load operation	1.05				ratio
bes_flex_price_capture	Ratio of realised to time-averaged wholesale electricity price for BES in flexible operation	1.3				ratio
capacity_factor_beccs	BECCS capacity factor	0.85				fraction
capacity_factor_bebcs	BEBCS capacity factor	0.8				fraction
discount_rate	Cost of capital (real, after-tax WACC for bioenergy projects)	0.056	0.068	0.062	0.071	fraction
capex_location_factor	Regional multiplier on CAPEX quoted on a US/EU basis (pyrolysis, BEBCS energy block, CO2 pipelines, liquefaction and terminals); not applied to BES/BECCS, whose CAPEX is set regionally	1.25	1.15	0.7	0.65	ratio
plant_om_factor	Annual O&M (fixed + variable) as a fraction of total CAPEX, for all conversion plants (BES, BECCS, pyrolysis, BEBCS energy block)	0.04				fraction
om_location_factor	Regional multiplier on the O&M-to-CAPEX fraction	1				ratio
haulage_location_factor	Regional multiplier on biomass haulage costs	1	1	0.75	0.65	ratio
elec_price	Wholesale electricity price received by the plant (scalar fallback; spatial runs use the regional price layer)	Spatially explicit	Spatially explicit	Spatially explicit	Spatially explicit	\$/MWh (2024 USD)
scaling_factor	Capital cost scale exponent	0.7				dimensionless
Spatial & Environmental						
ff_c_intensity	Displaced grid carbon intensity at today's effective carbon price (empirical build margin, IPCC life-cycle, biomass = 0)	Spatially explicit	Spatially explicit	Spatially explicit	Spatially explicit	tCO2eq/GJ

Table 1: Operational, financial, and biophysical parameters of the C-SCAPE model.

Parameter	Description	Value				Units
		USA/Default	Europe	China	India	
grid_p_now	Current explicit carbon price on power-sector emissions (MEF(P) anchor)	0	70	14	0	\$/tCO ₂ (2024 USD)
mef_floor	Residual life-cycle intensity of a decarbonised grid (MEF(P) floor)	0.02				tCO ₂ /MWh
mef_draw_u	Selects a bootstrap draw of the MEF(P) curve (0 = central fit)	0				fraction
soil_cec	Cation Exchange Capacity	20				cmol / kg
soil_ph	Native topsoil pH	Spatially explicit	Spatially explicit	Spatially explicit	Spatially explicit	pH
soil_temp	Mean annual soil temperature	14.9				C
Technology & Thermodynamics						
plant_mw_th	Target plant thermal capacity	125				MWth
Technology - BECCS						
beccs_capex_premium	Increase in total CAPEX for post-combustion capture and compression relative to BES at equal thermal input	0.4				fraction
beccs_eff_penalty	BECCS efficiency penalty (percentage points below BES)	0.08				fraction
capture_rate	BECCS CO ₂ capture efficiency	0.9				fraction
Technology - BES						
bes_capital_cost	Local CAPEX for BES at a 50 MWe reference plant (quoted at bes_capex_ref_eff); already regional, so capex_location_factor is not applied	4700	4300	2200	1900	\$/kWe (net) (2024 USD)
bes_capex_ref_eff	Net electrical efficiency at which bes_capital_cost is quoted; sizes combustion CAPEX from thermal input	0.3				fraction
bes_energy_efficiency	Electrical efficiency for BES	0.3				fraction

Table 1: Operational, financial, and biophysical parameters of the C-SCAPE model.

Parameter	Description	Value				Units
		USA/Default	Europe	China	India	
bes_life	BES (and BECCS) plant lifetime	25				years
Technology - Pyrolysis (BEBCS)						
py_cc	Pyrolysis unit capital cost at the base location (scaled by <code>capex_location_factor</code>), per Mg/yr of feed at the 50 MWe reference size	268				\$/ (Mg feed/yr) (2024 USD)
py_life	Pyrolysis project lifetime	20				years
bebc power_life	BEBCS power block lifetime	25				years
bebc power_capital_cost	Pyrolysis vapour/gas combustion and steam power block capital cost at the base location (scaled by <code>capex_location_factor</code>)	2600				\$/kWe (2024 USD)
py_temp	Peak pyrolysis temperature	500				°C
py_heater_eff	Pyrolysis process heater efficiency (HHV basis)	0.8				fraction
py_parasitic_power	Pyrolysis plant parasitic electricity demand	0.07				GJe / Mg feed
py_exhaust_temp	Temperature of vapours and gases leaving heat recovery	170				°C
bebc power_efficiency	Electrical efficiency of pyrolysis fuel power block (LHV basis)	0.35				fraction

5 FIGURES

6 REFERENCES AND NOTES

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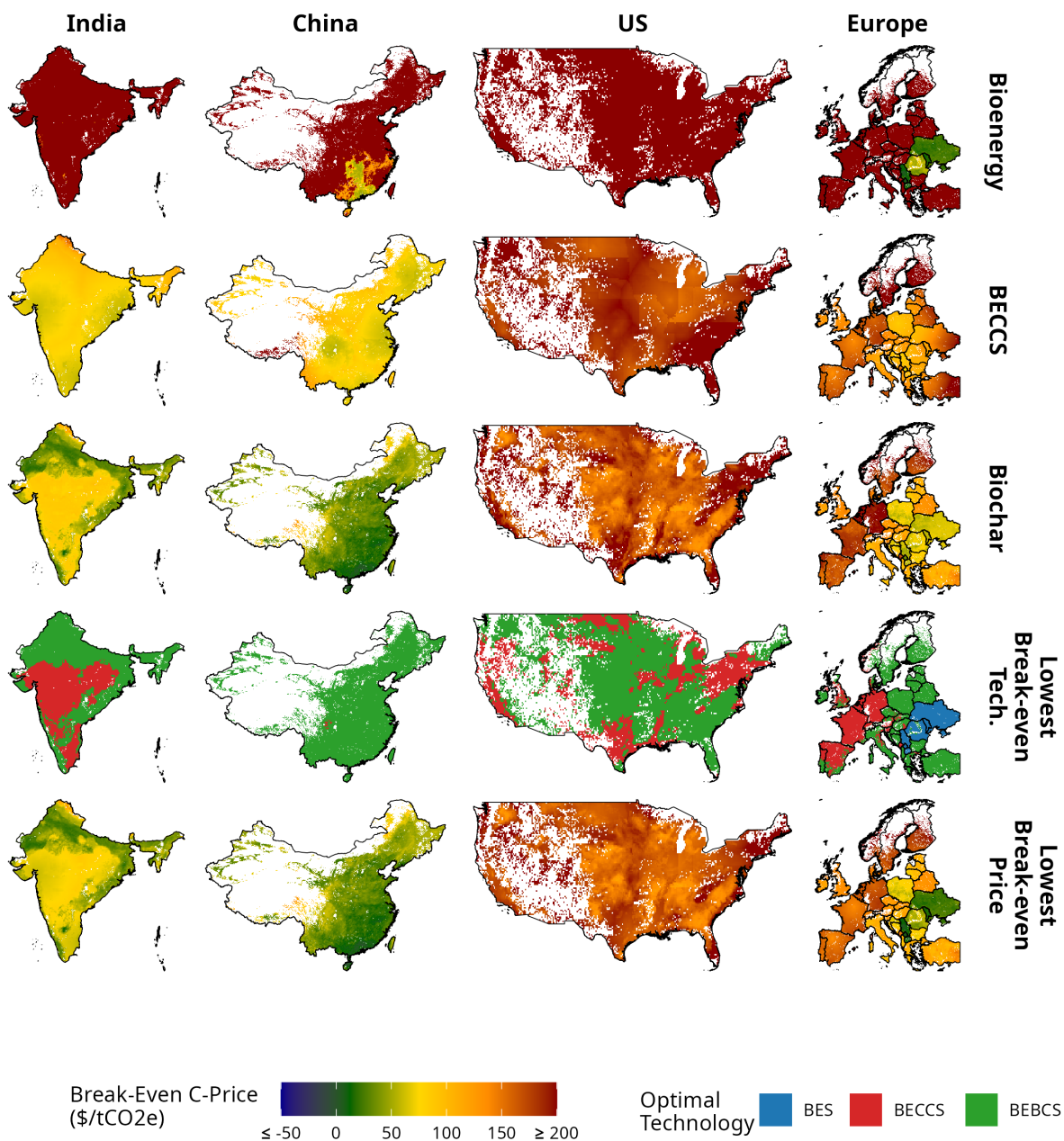


Figure 1: Minimum Break-even carbon price for each technology option. The penultimate row shows the technology with the lowest break-even price, and the final row shows that price. The lowest break-even technology is the first to become profitable as the carbon price rises; it is not necessarily the technology with the highest NPV at a given carbon price (see Figure 2 to Figure 5).

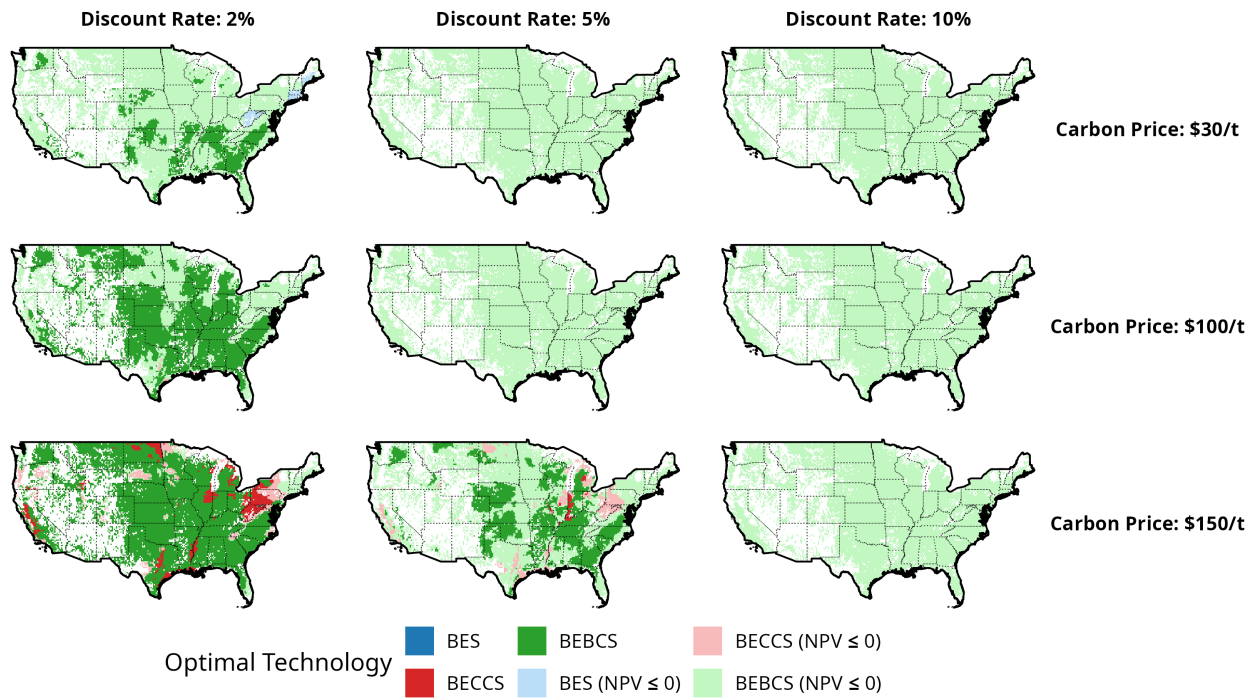


Figure 2: Optimal technology as a function of carbon price and discount rate (USA).

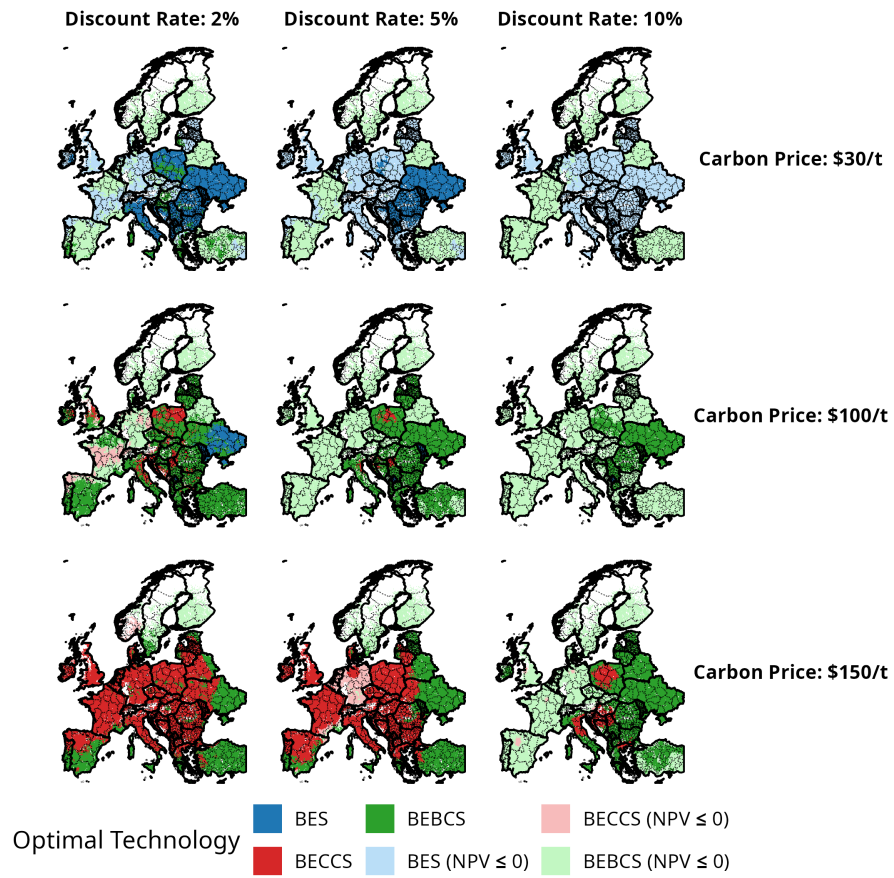


Figure 3: Optimal technology as a function of carbon price and discount rate (Europe).

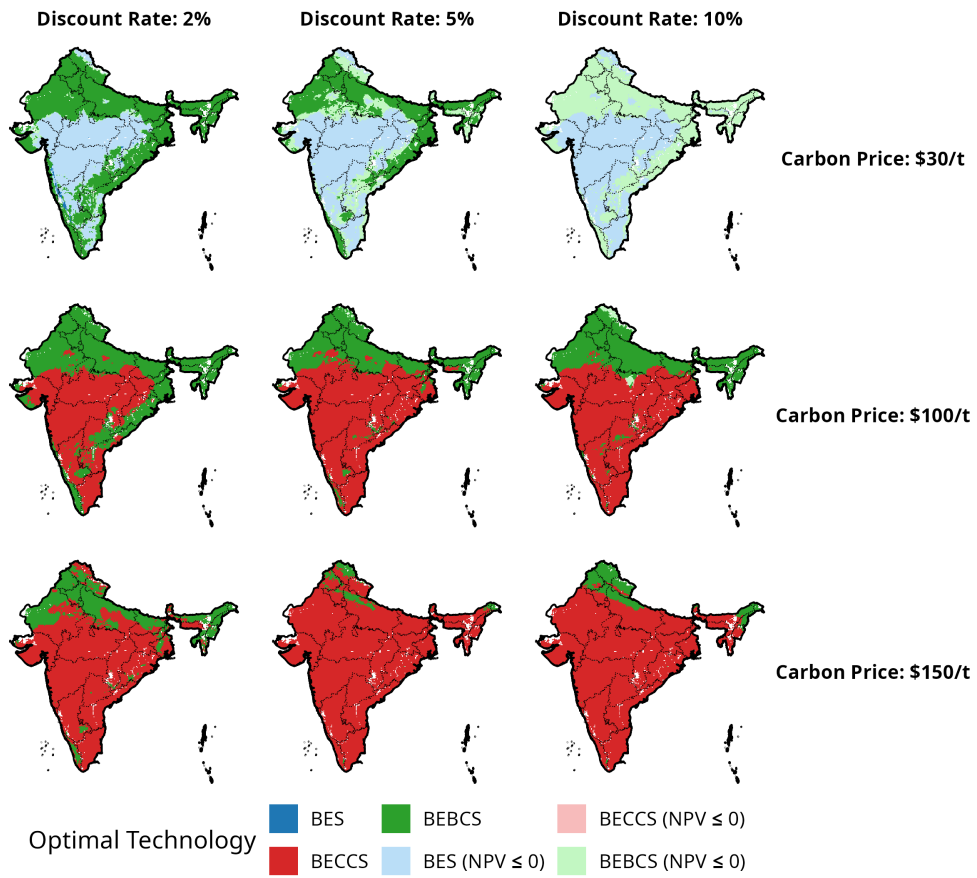


Figure 4: Optimal technology as a function of carbon price and discount rate (India).

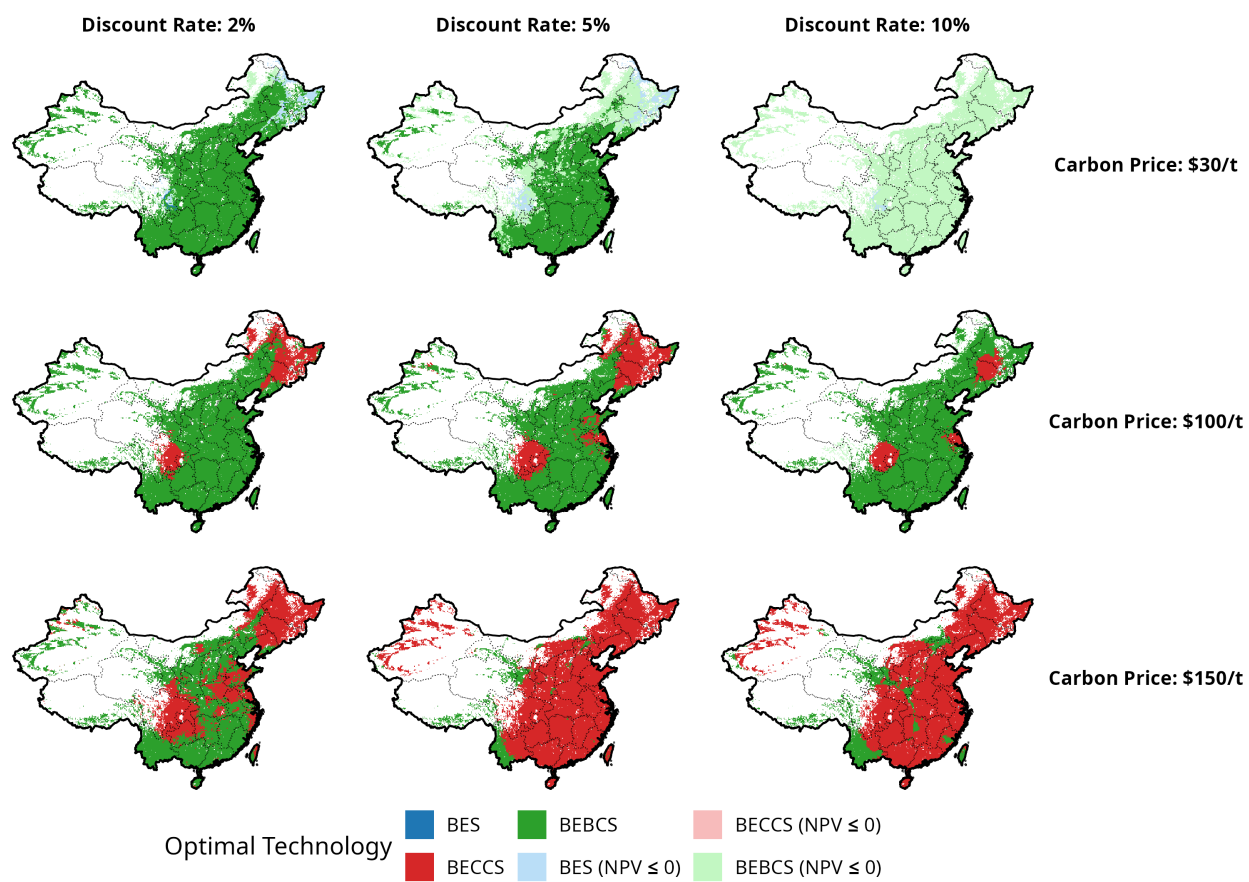


Figure 5: Optimal technology as a function of carbon price and discount rate (China).

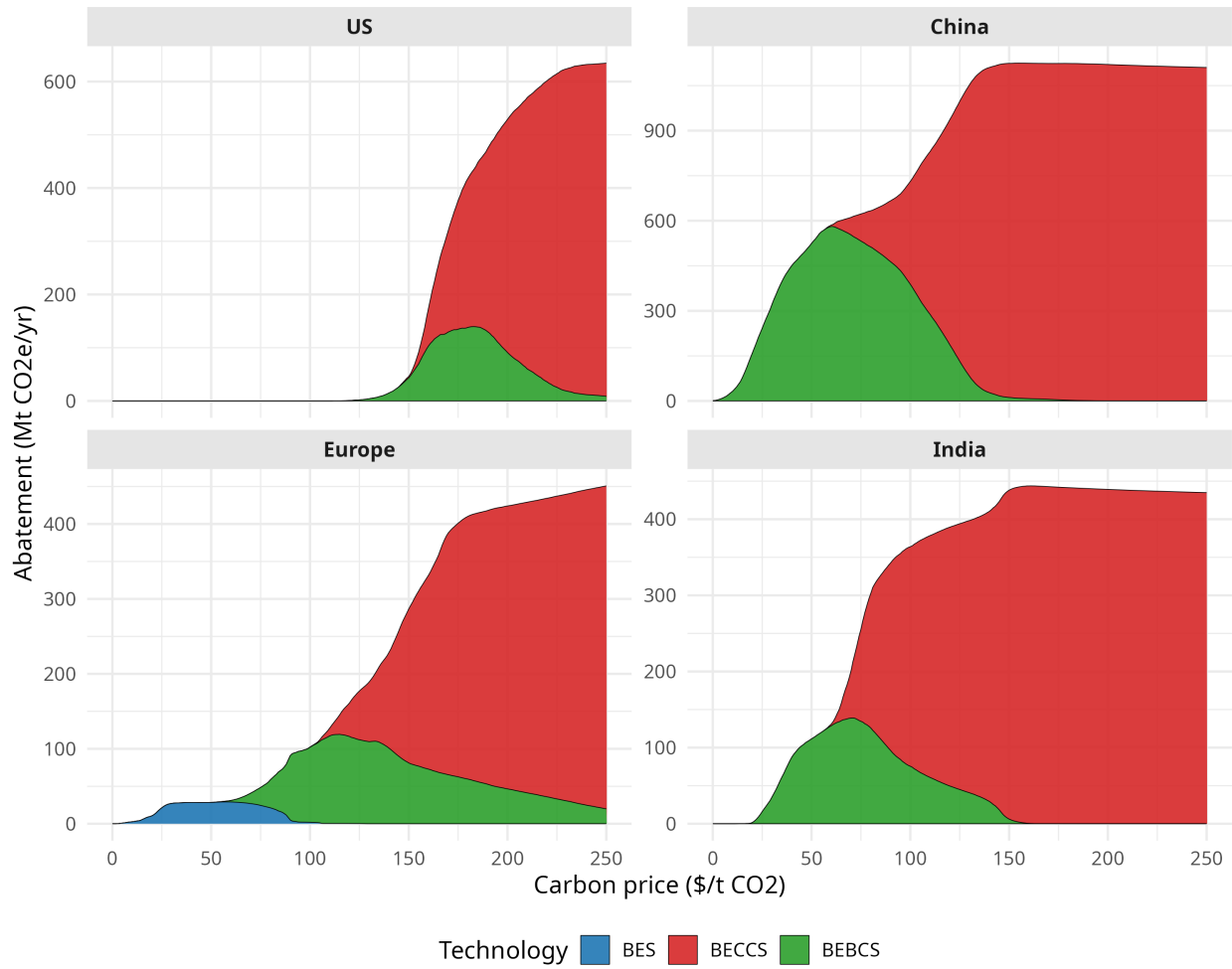


Figure 6: Marginal abatement cost curves: abatement potential of the technology with the highest non-negative net value in each grid cell, by region, as a function of the carbon price.

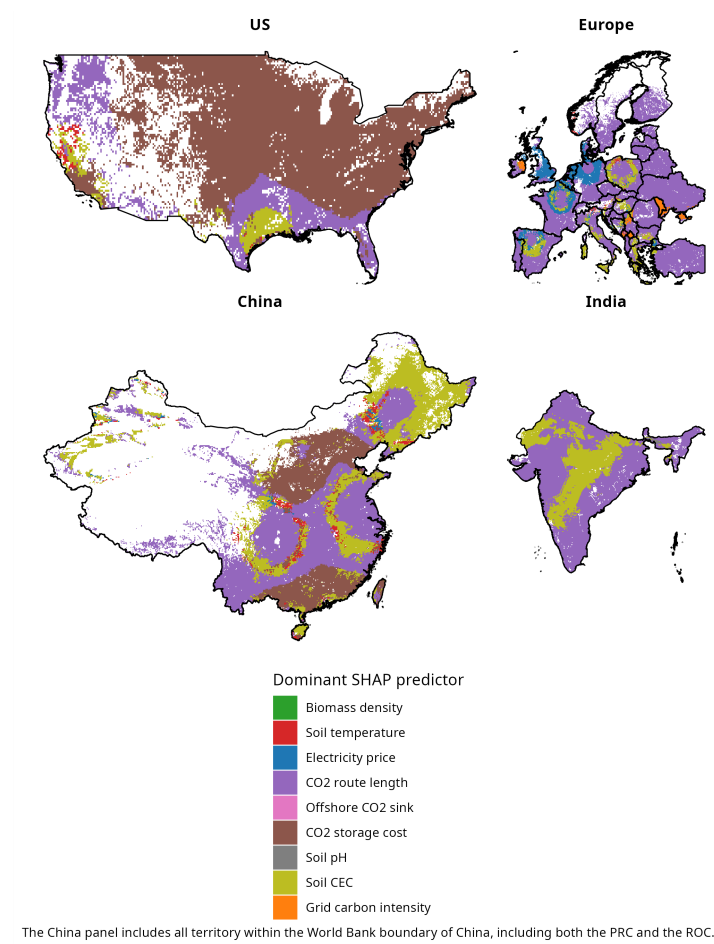


Figure 7: Dominant spatial driver of the optimal technology in each grid cell: the input with the largest absolute SHAP value for the cell's optimal technology, from surrogate classifiers fitted separately for each region (scenario CP100_MW250; see Supplementary Information). The China panel includes all territory within the World Bank boundary of China, including both the PRC and the ROC.

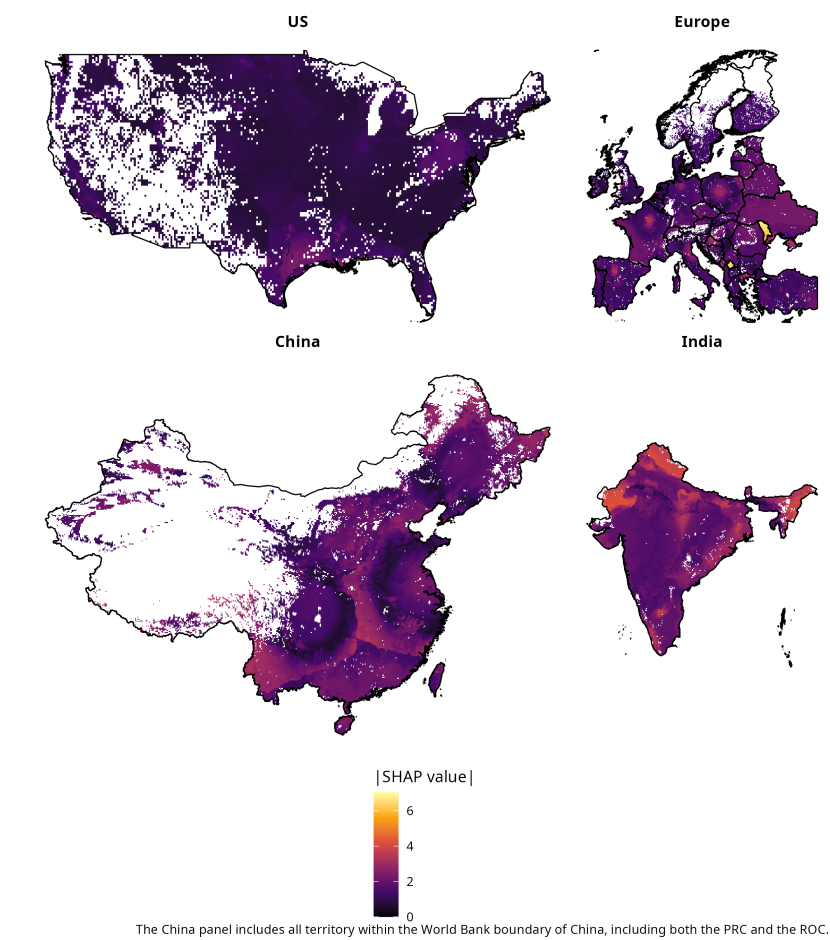


Figure 8: Magnitude (absolute SHAP value, log-odds) of the dominant spatial driver in each grid cell, from the regional surrogate classifiers. The China panel includes all territory within the World Bank boundary of China, including both the PRC and the ROC.

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898 **SUPPLEMENTARY MATERIALS**

Supplementary tables

Table S1: Parameters sampled in the Monte Carlo analysis: central values (by region where they differ), distribution and range. Generated from BiocharAG/inst/extdata/parameters.csv.

Parameter	Description	Units	Central	Dist.	Range
Biochar & Agronomic					
ash_cce	Calcium carbonate equivalent of recycled combustion ash	fraction	0.85	Pert	0.4–1
ash_p_content	Phosphorus content of recycled combustion ash (elemental P)	fraction	0.012	Pert	0.5–2 × central
bc_cce	Calcium Carbonate Equivalent	fraction	0.15	Pert	0.05–0.3
bc_cec_value	Annual crop-yield benefit of biochar from raising soil CEC, at CEC ≤ 5 cmol/kg, falling linearly to 0 at CEC ≥ 30; valued as a perpetuity, annual value / (discount rate + biochar C decay rate)	\$/Mg biochar C/yr (2024 USD)	21.1	Pert	0.5–1.5 × central
bc_field_cost	Biochar field application cost (spreading and incorporation)	\$/ha (2024 USD)	116	Pert	0–233
bc_field_diesel	Tractor diesel use for biochar spreading and incorporation	L/ha	17.5	Pert	13–30
bc_k_content	Biochar potassium content (elemental K)	fraction	0.005	Pert	0.5–2 × central
bc_n_content	Biochar Nitrogen content	fraction	0.005	Pert	0.5–2 × central
bc_p_content	Biochar phosphorus content (elemental P)	fraction	0.002	Pert	0.5–2 × central
n2o_ef	Direct N2O emission factor for fertiliser N	kg N2O-N / kg N	0.01	Pert	0.001–0.018
n2o_reduction	Fractional reduction in soil N2O emissions after biochar application	fraction	0.23	Pert	0.05–0.41
n2o_years	Duration of biochar N2O reduction	years	2	Pert	1–15
n_app_rate	Mineral N fertiliser application rate on cropland	kg N/ha/yr	US 60; Europe 74; China 198; India 118	Pert	0.75–1.25 × central
price_k	Market price for potassium (world market; all regions)	\$/kg K2O (2024 USD)	0.86	Pert	0.6–1.8 × central

(continued)

Parameter	Description	Units	Central	Dist.	Range
<code>price_lime</code>	Cost of agricultural lime delivered and spread, per Mg of calcium carbonate equivalent (100% ECCE)	\$/Mg CaCO ₃ -eq (2024 USD)	59	Pert	31–110
<code>price_n</code>	Market price for nitrogen (world market; all regions)	\$/kg N (2024 USD)	1.06	Pert	0.6–1.8 × central
<code>price_p</code>	Market price for phosphorus (world market; all regions)	\$/kg P ₂ O ₅ (2024 USD)	1.27	Pert	0.6–1.8 × central
CO₂ Transport & Storage					
<code>ccs_storage_cost</code>	Onshore injection and monitoring (exploration to post-closure), for sinks without a site-specific cost	\$/Mg CO ₂ (2024 USD)	10	Pert	2–33
<code>co2_ship_emis_fixed</code>	CO ₂ emitted per tonne shipped, fixed part (liquefaction, loading, intermediate storage)	t CO ₂ / t CO ₂	0.022	Pert	0.5–1.5 × central
<code>co2_ship_emis_per_km</code>	CO ₂ emitted per tonne shipped per km of voyage	t CO ₂ / t CO ₂ / km	0.000013	Pert	0.5–1.5 × central
<code>co2_subsea_capex_factor</code>	Subsea CO ₂ pipeline CAPEX per km relative to onshore (offshore sinks reached by pipeline)	ratio	1.5	Pert	1.3–2
<code>cost_offshore_storage</code>	Offshore injection and monitoring (exploration to post-closure), for sinks without a site-specific cost	\$/Mg CO ₂ (2024 USD)	20	Pert	5–50
<code>storage_cost_factor</code>	Multiplier on site-specific CO ₂ storage costs (classified sinks in <code>co2_sinks</code>)	ratio	1	Pert	0.5–1.75
Feedstock & Logistics					
<code>bm_ash</code>	Biomass ash mass fraction	fraction	US 0.05; Europe 0.05; China 0.15; India 0.15	Pert	0.6–1.6 × central
<code>bm_c</code>	Biomass carbon mass fraction	fraction	0.48	Pert	0.92–1.04 × central
<code>bm_h2o</code>	Biomass moisture content	fraction	0.1	Pert	0.08–0.2
<code>bm_lhv</code>	Biomass lower heating value (dry ash-free)	GJ/Mg daf	18.6	Pert	0.95–1.05 × central
<code>bm_transport_fixed</code>	Fixed loading/handling cost	\$/Mg (2024 USD)	6.27	Pert	0.7–1.5 × central

(continued)

Parameter	Description	Units	Central	Dist.	Range
<code>bm_transport_var</code>	Variable trucking cost per km of one-way haul (includes the empty return)	\$/Mg/km (2024 USD)	0.19	Pert	$0.7\text{--}1.5 \times \text{central}$
<code>feedstock_storage_cost</code>	Interim storage of seasonally harvested residues for year-round supply (about six months), US/EU basis; scaled by <code>haulage_location_factor</code>	\$/Mg (2024 USD)	15	Pert	3–42
<code>haul_fuel_share</code>	Share of variable trucking cost that is fuel	fraction	0.24	Pert	0.18–0.32
<code>haul_time_share</code>	Share of variable trucking cost that is time-based (driver wages and benefits; truck payments; insurance; permits)	fraction	0.64	Pert	0.55–0.72
<code>india_feedstock_cost</code>	Field-side price of baled paddy straw in India, dry basis (used for India only; other regions use <code>calculate_regional_feedstock_cost()</code> values)	\$/Mg (2024 USD)	34	Pert	12–50
<code>lignin</code>	Lignin fraction	fraction	0.2	Pert	$0.75\text{--}1.25 \times \text{central}$
<code>residue_burn_factor</code>	Multiplier on the cell-level burnt share (<code>residue_burn_map</code>)	ratio	1	Pert	0.5–1.5
Financial & Market					
<code>bes_base_price_capture</code>	Ratio of realised to time-averaged wholesale electricity price for BES in base-load operation	ratio	1.05	Pert	1.02–1.08
<code>bes_flex_price_capture</code>	Ratio of realised to time-averaged wholesale electricity price for BES in flexible operation	ratio	1.3	Pert	1.15–1.45
<code>capacity_factor_bebcs</code>	BEBCS capacity factor	fraction	0.8	Pert	0.7–0.85
<code>capacity_factor_beccs</code>	BECCS capacity factor	fraction	0.85	Pert	0.6–0.9
<code>capacity_factor_bes_base</code>	BES capacity factor in base-load operation	fraction	0.85	Pert	0.75–0.9
	BES capacity factor in flexible (load-following) operation	fraction	0.45	Pert	0.3–0.6
<code>capacity_factor_bes_flex</code>					

(continued)

Parameter	Description	Units	Central	Dist.	Range
<code>elec_price</code>	Wholesale electricity price received by the plant (scalar fallback; spatial runs use the regional price layer)	\$/MWh (2024 USD)	60 (spatial)	Pert	$0.7\text{--}1.5 \times \text{central}$
<code>plant_om_factor</code>	Annual O&M (fixed + variable) as a fraction of total CAPEX, for all conversion plants (BES, BECCS, pyrolysis, BEBCS energy block)	fraction	0.04	Pert	$0.75\text{--}1.35 \times \text{central}$
Spatial & Environmental					
<code>ff_c_intensity</code>	Displaced grid carbon intensity at today's effective carbon price (empirical build margin, IPCC life-cycle, biomass = 0)	tCO ₂ eq/GJ	0.003333 (spatial)	Pert	$0.8\text{--}1.2 \times \text{central}$
<code>mef_draw_u</code>	Selects a bootstrap draw of the MEF(P) curve (0 = central fit)	fraction	0	Uniform	0–1
Technology - BECCS					
<code>beccs_capex_premium</code>	Increase in total CAPEX for post-combustion capture and compression relative to BES at equal thermal input	fraction	0.4	Pert	0.25–0.8
<code>beccs_eff_penalty</code>	BECCS efficiency penalty (percentage points below BES)	fraction	0.08	Pert	0.05–0.13
<code>capture_rate</code>	BECCS CO ₂ capture efficiency	fraction	0.9	Pert	0.85–0.95
Technology - BES					
<code>bes_capital_cost</code>	Local CAPEX for BES at a 50 MWe reference plant (quoted at <code>bes_capex_ref_eff</code>); already regional, so <code>capex_location_factor</code> is not applied	\$/kWe (net) (2024 USD)	US 4700; Europe 4300; China 2200; India 1900	Pert	$0.85\text{--}1.25 \times \text{central}$ (India 0.5–1.25)
<code>bes_energy_efficiency</code>	Electrical efficiency for BES	fraction	0.3	Pert	$0.85\text{--}1.15 \times \text{central}$
Technology - Pyrolysis (BEBCS)					
<code>bebc_power_capital_cost</code>	Pyrolysis vapour/gas combustion and steam power block capital cost at the base location (scaled by <code>capex_location_factor</code>)	\$/kWe (2024 USD)	2600	Pert	$0.8\text{--}1.25 \times \text{central}$

(continued)

Parameter	Description	Units	Central	Dist.	Range
py_cc	Pyrolysis unit capital cost at the base location (scaled by capex_location_factor), per Mg/yr of feed at the 50 MWe reference size	\$(Mg feed/yr) (2024 USD)	268	Pert	$0.7\text{--}1.8 \times \text{central}$
py_heater_eff	Pyrolysis process heater efficiency (HHV basis)	fraction	0.8	Pert	0.7–0.9
py_parasitic_power	Pyrolysis plant parasitic electricity demand	GJe / Mg feed	0.07	Pert	$0.5\text{--}2 \times \text{central}$
py_temp	Peak pyrolysis temperature	°C	500	Pert	$0.9\text{--}1.1 \times \text{central}$

Table S2: Share of mean absolute SHAP value (% , summed over classes) by input and model, and the training accuracy of each classifier. A dash marks an input that is constant within the region and was dropped from that region’s model.

Input	Global	US	Europe	China	India
Electricity price	26.1	0.3	11.2	6.7	–
CO2 route length	16.0	21.9	32.9	31.6	56.9
Feedstock cost	15.7	–	0.2	–	–
Soil CEC	14.0	21.4	13.3	27.0	36.9
Grid carbon intensity	10.4	0.1	20.9	–	–
Soil temperature	9.0	21.9	14.0	9.6	2.9
CO2 storage cost	6.6	30.5	3.7	19.8	0.0
Biomass density	1.1	2.4	1.2	1.4	0.4
Soil pH	0.9	1.6	0.7	3.8	2.8
Offshore CO2 sink	0.3	0.0	1.9	0.2	0.1
Training accuracy	99.0	100.0	99.6	99.2	99.9

900 Supplementary figures

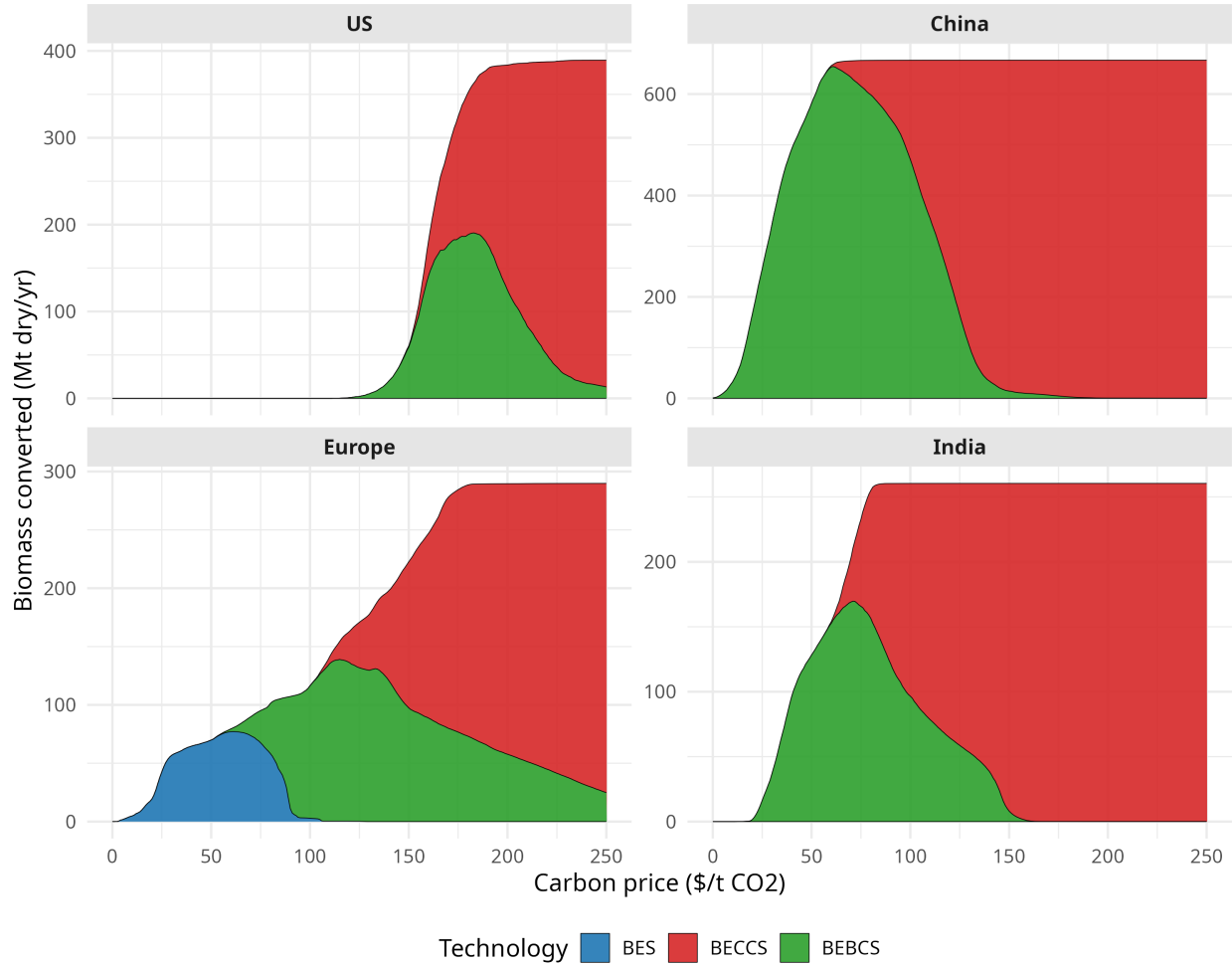


Figure S1: Biomass converted by the technology with the highest non-negative net value in each grid cell, by region, as a function of the carbon price (the biomass equivalent of Figure 6).

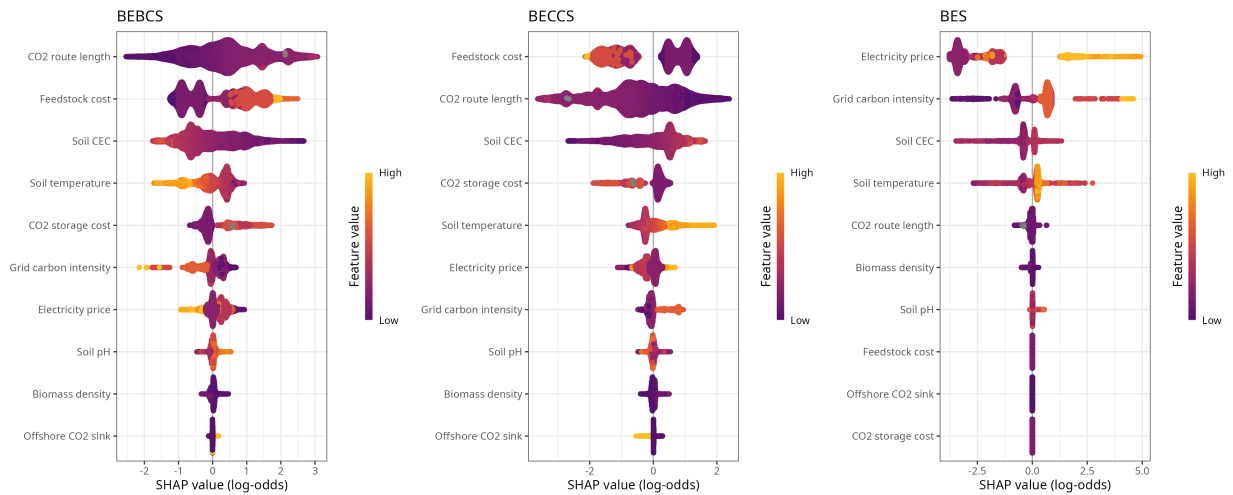


Figure S2: SHAP values of each input for each optimal technology, global model (all regions). Each point is a grid cell; colour shows the input value (low to high).

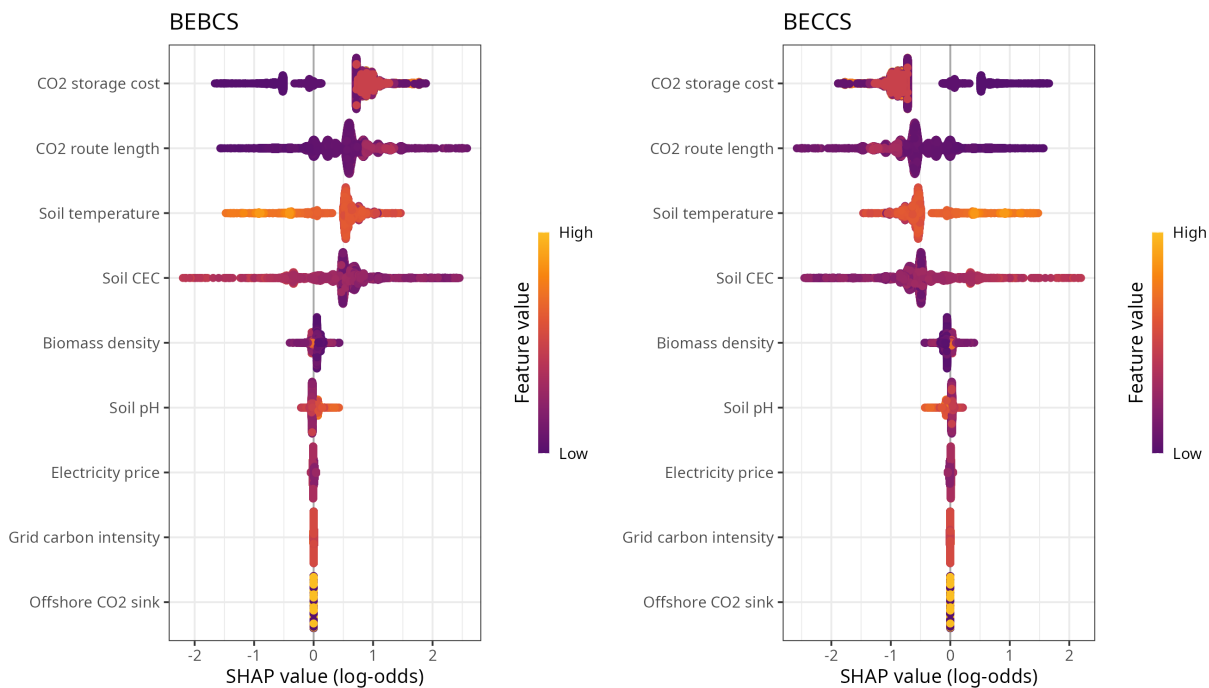


Figure S3: SHAP values by input, United States model.

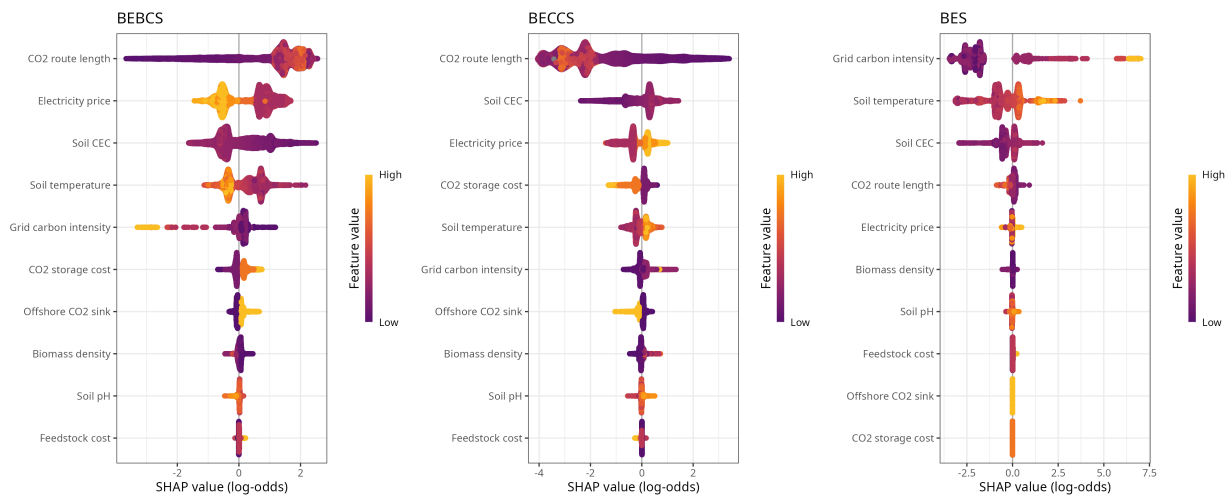


Figure S4: SHAP values by input, Europe model.

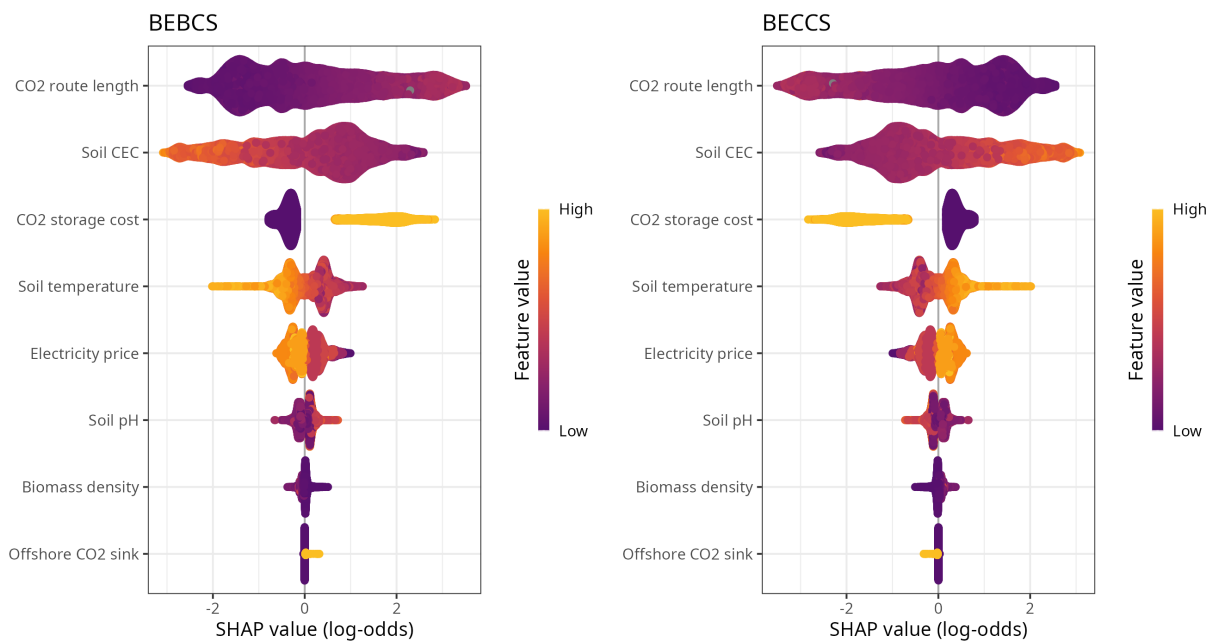


Figure S5: SHAP values by input, China model.

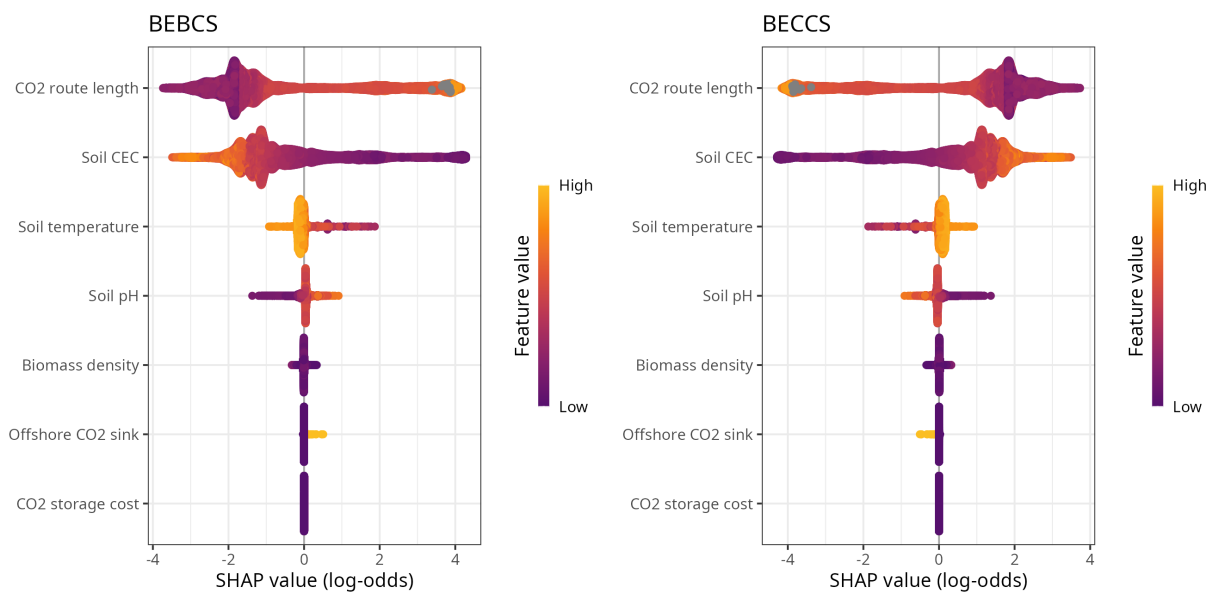


Figure S6: SHAP values by input, India model.

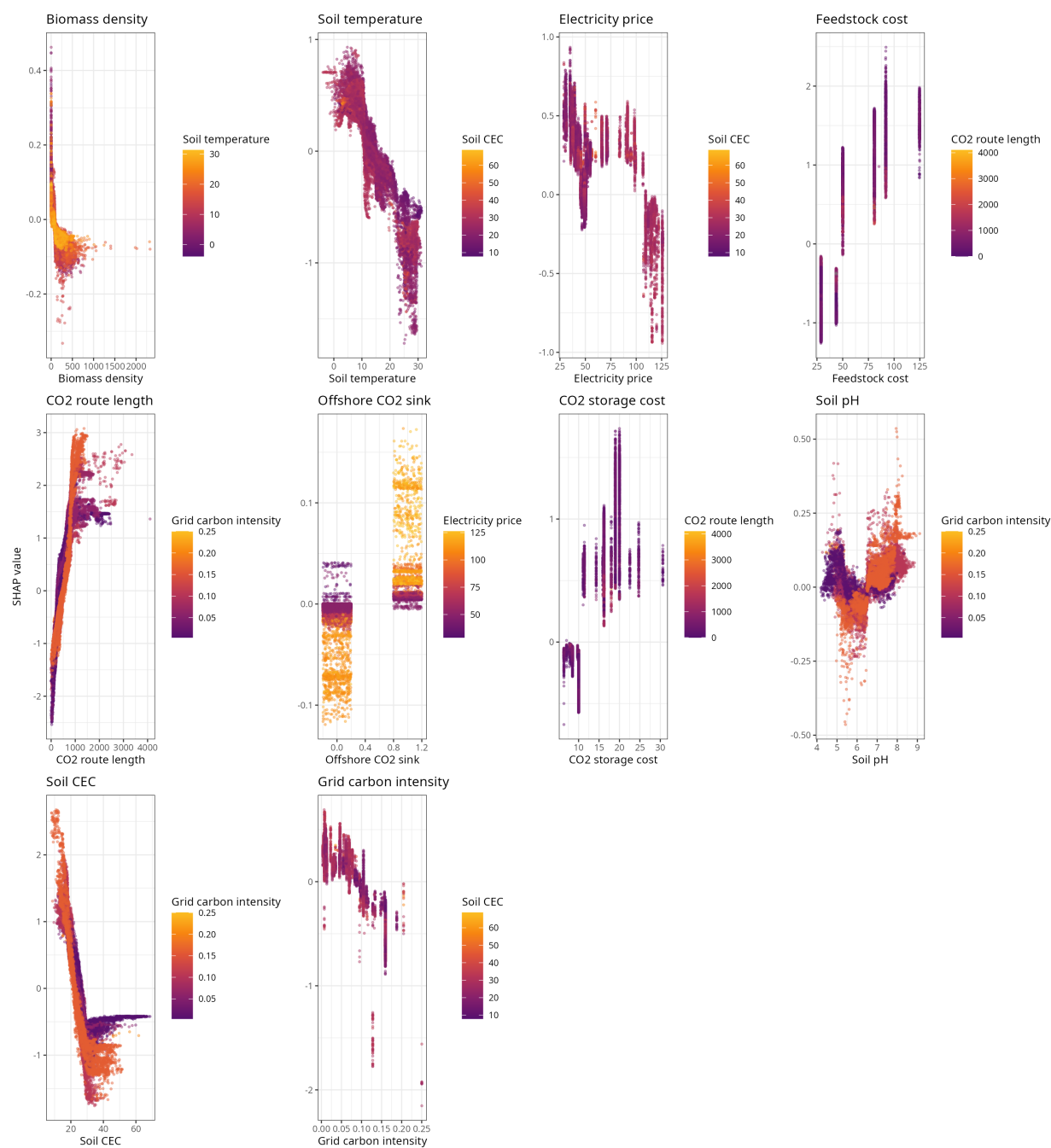


Figure S7: SHAP dependence plots for BEBCS, global model (all regions): SHAP value of each input against its value, coloured by the input with the strongest interaction.

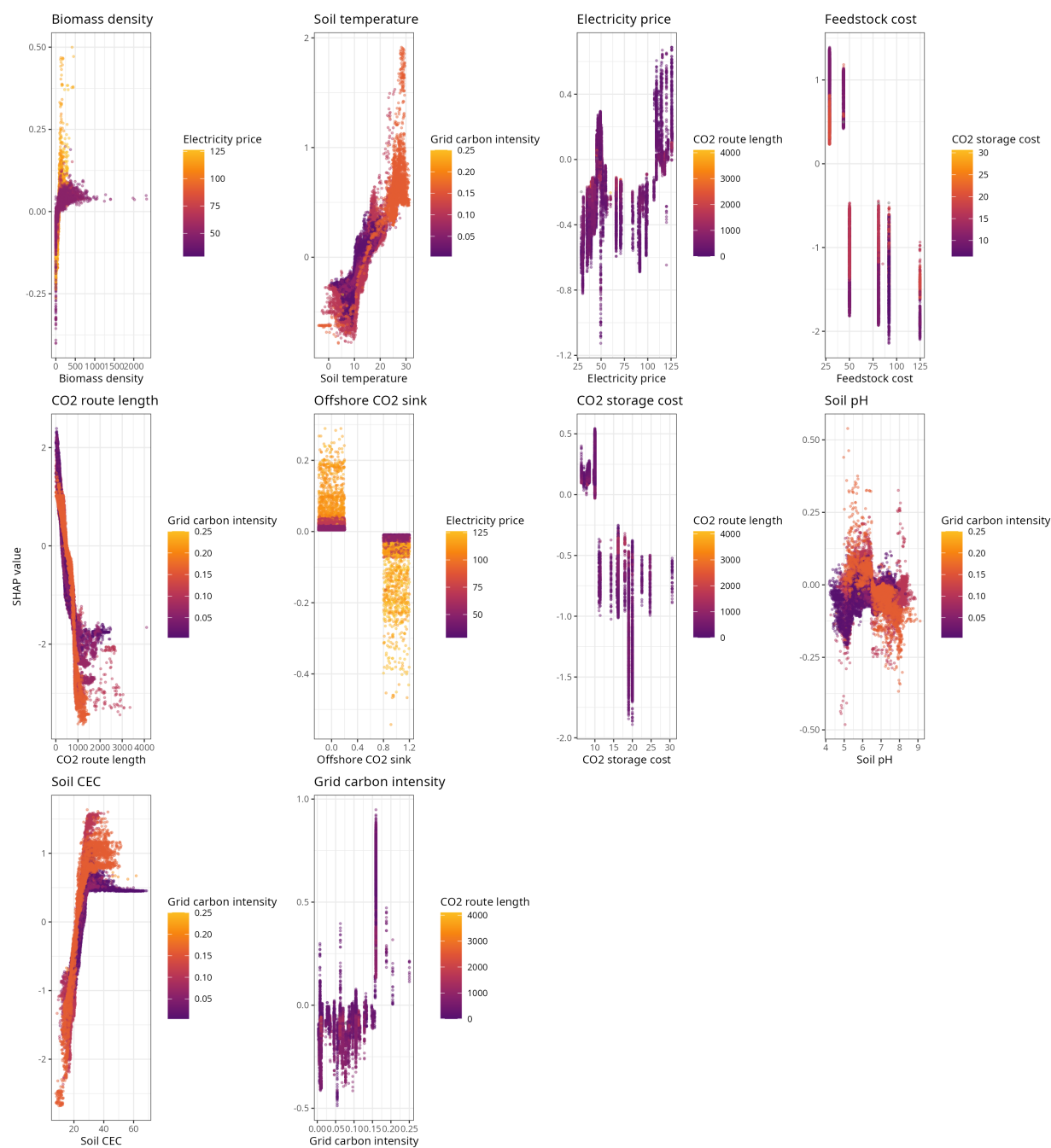


Figure S8: SHAP dependence plots for BECCS, global model (all regions): SHAP value of each input against its value, coloured by the input with the strongest interaction.

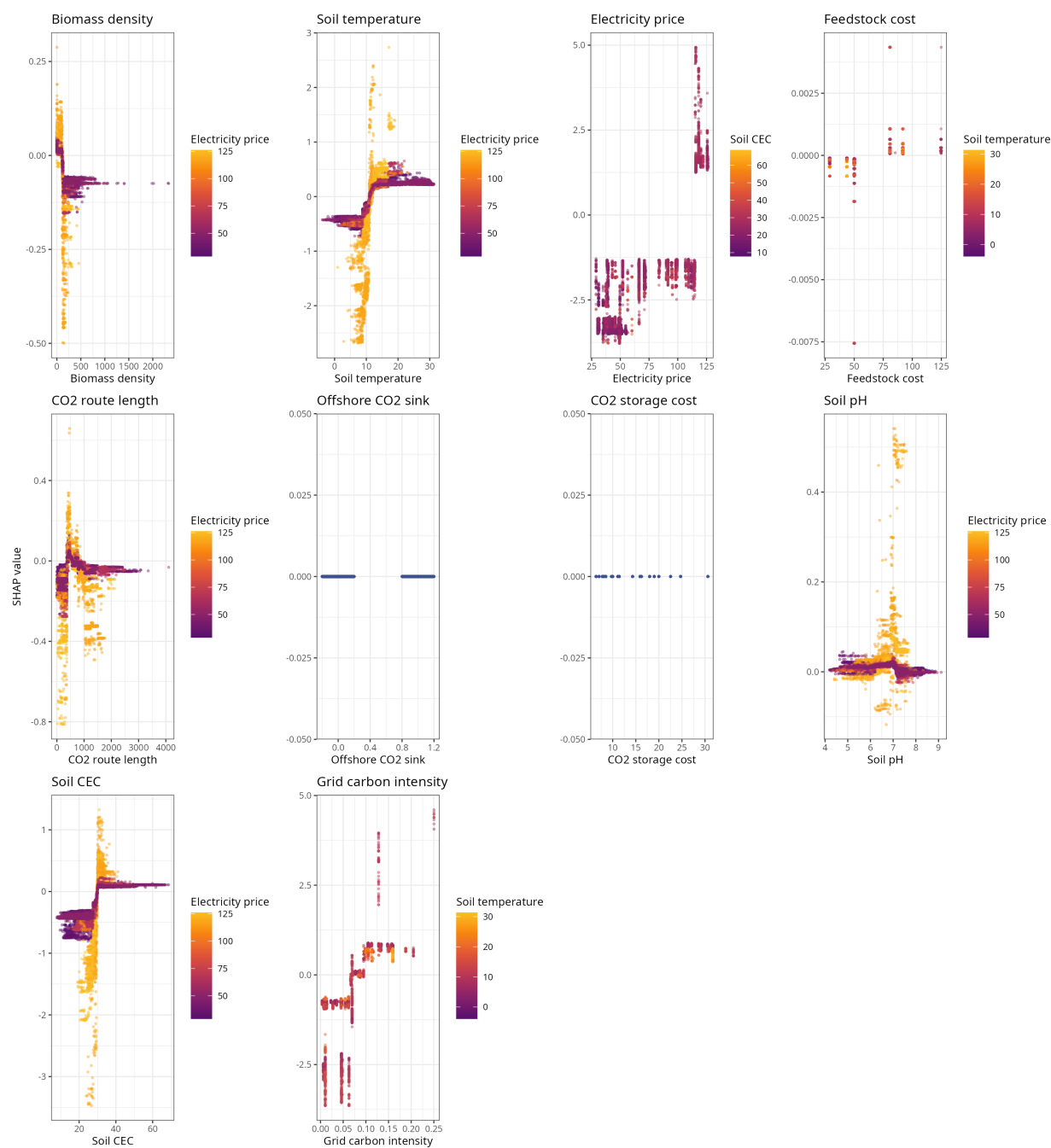


Figure S9: SHAP dependence plots for BES, global model (all regions): SHAP value of each input against its value, coloured by the input with the strongest interaction.

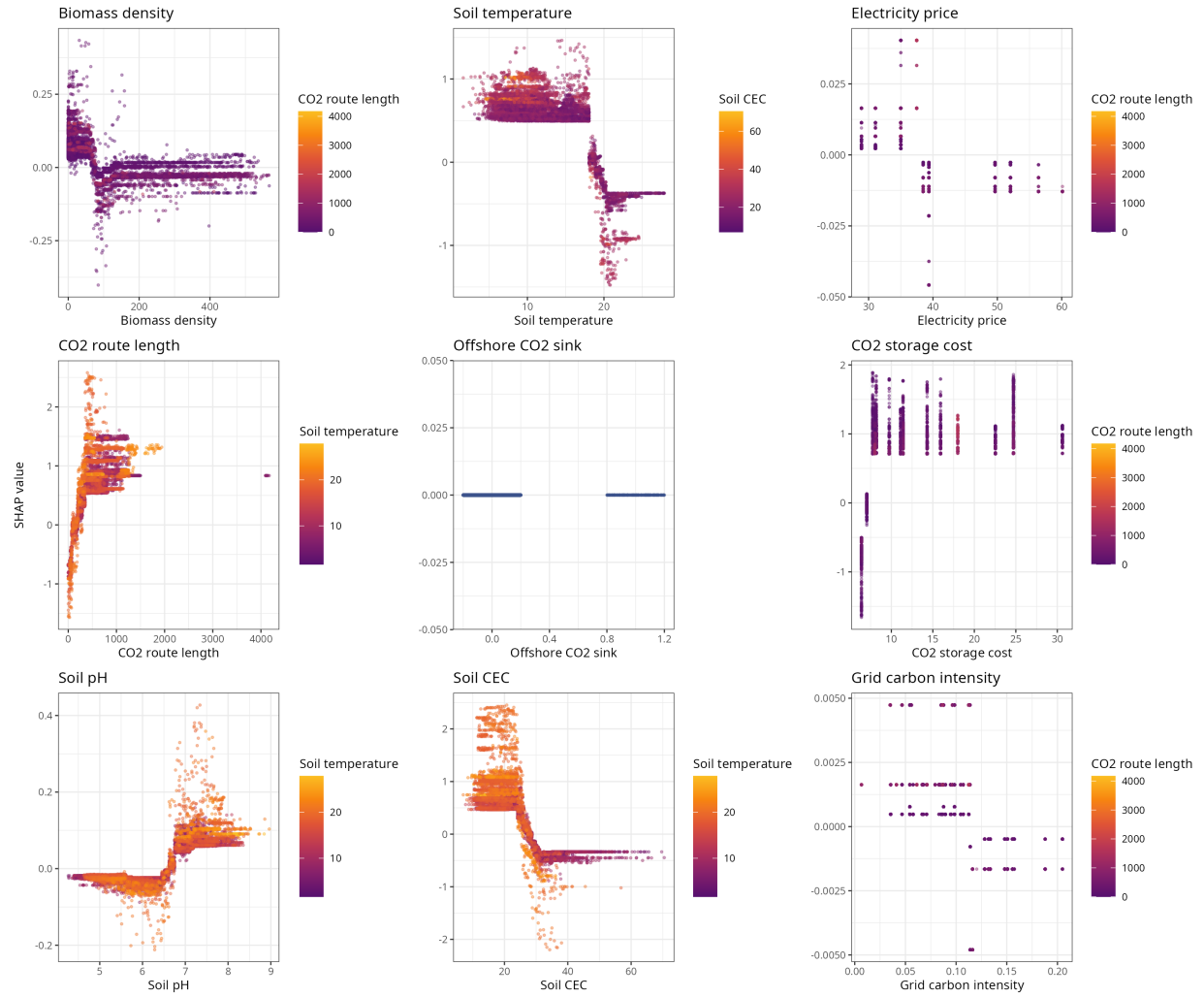


Figure S10: SHAP dependence plots for BEBCS, United States model: SHAP value of each input against its value, coloured by the input with the strongest interaction.

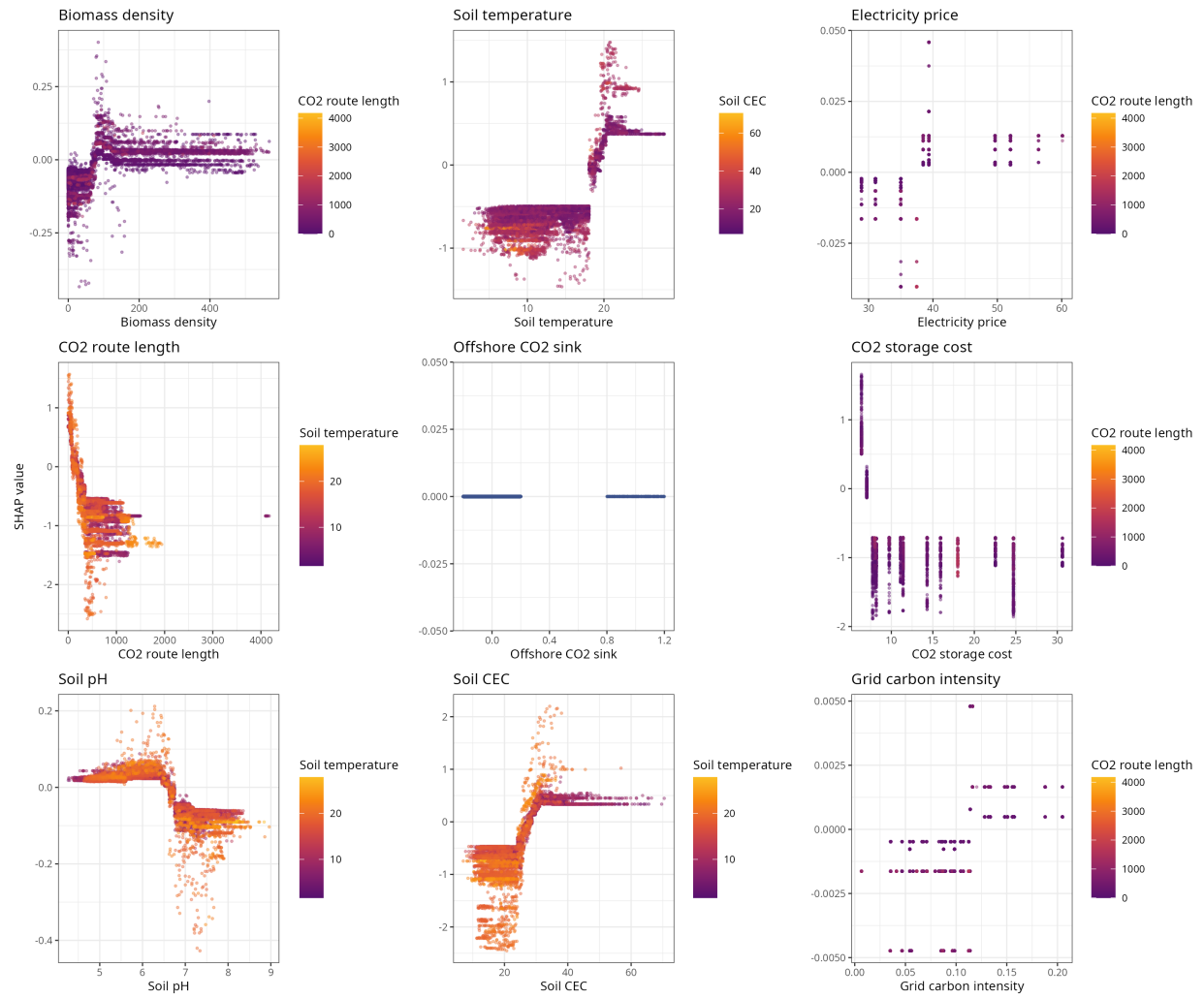


Figure S11: SHAP dependence plots for BECCS, United States model: SHAP value of each input against its value, coloured by the input with the strongest interaction.

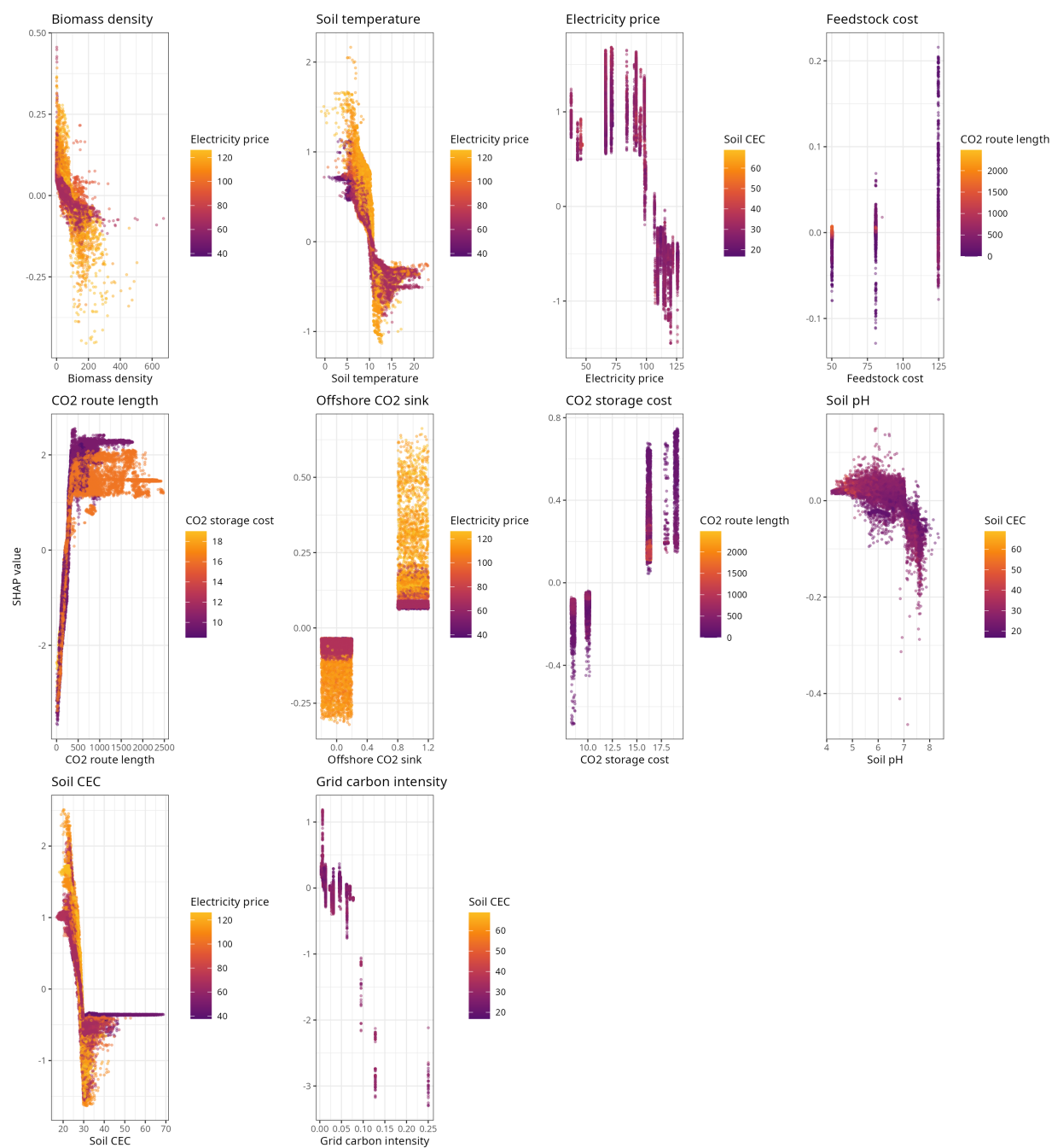


Figure S12: SHAP dependence plots for BEBCS, Europe model: SHAP value of each input against its value, coloured by the input with the strongest interaction.

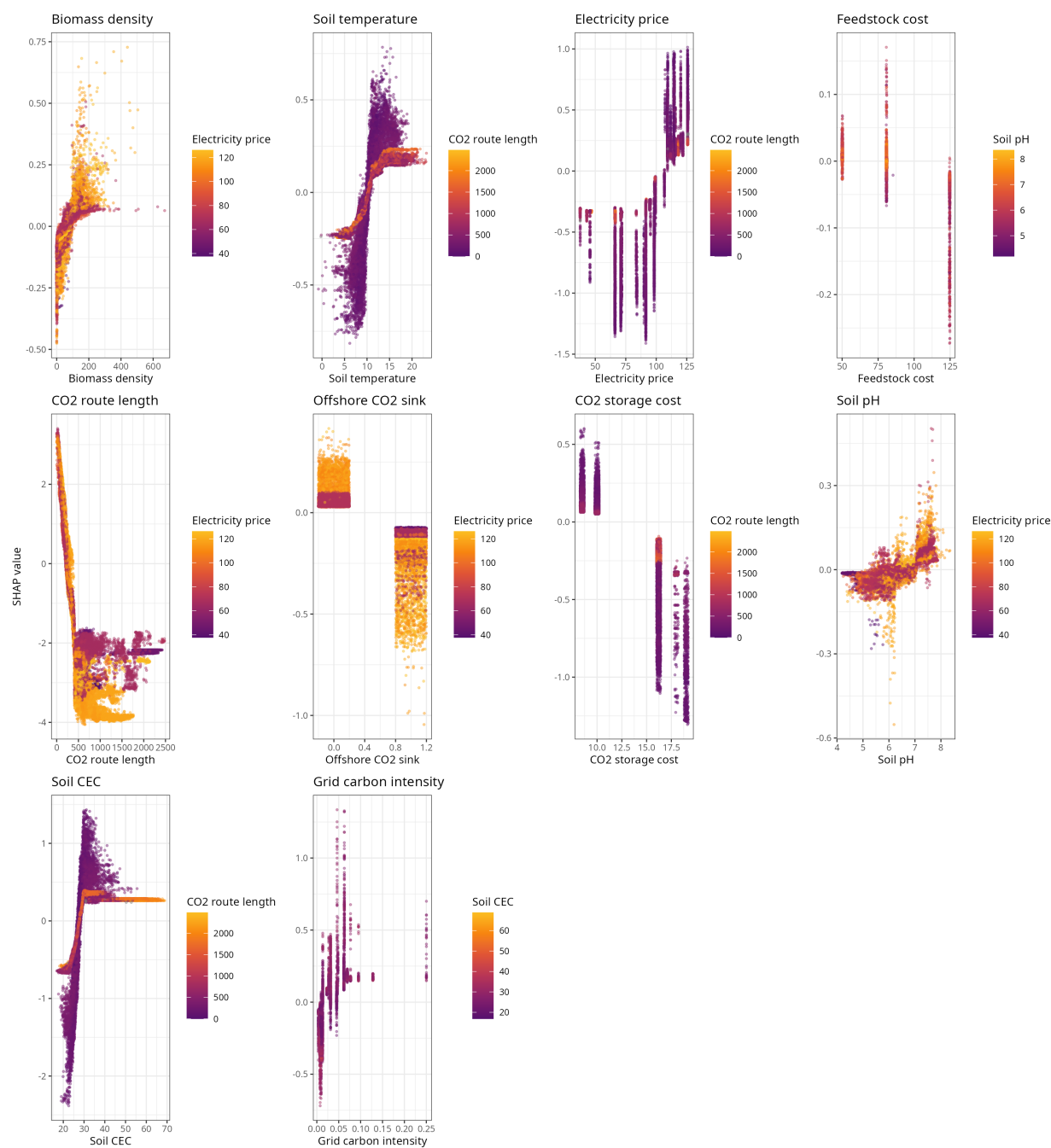


Figure S13: SHAP dependence plots for BECCS, Europe model: SHAP value of each input against its value, coloured by the input with the strongest interaction.

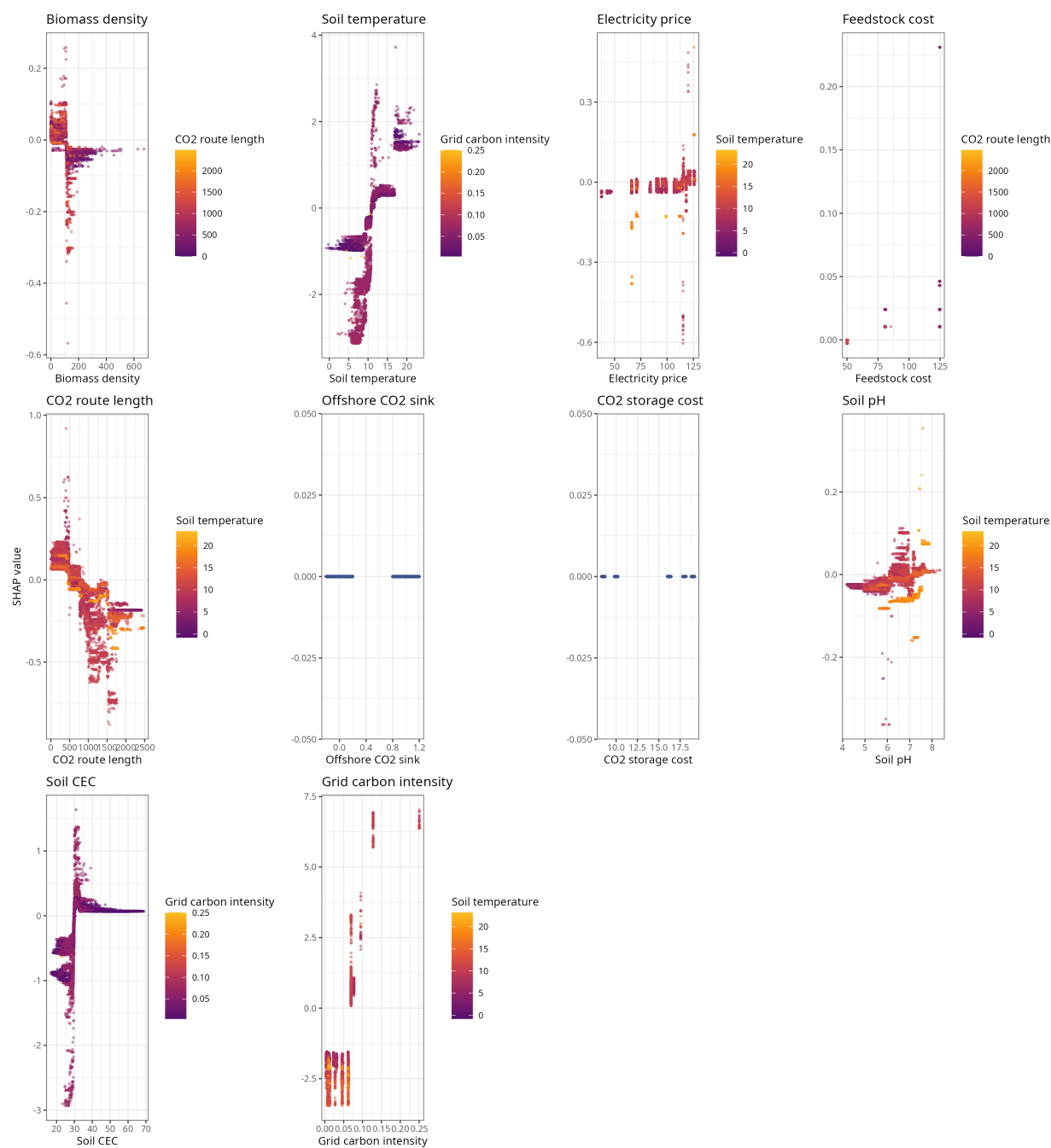


Figure S14: SHAP dependence plots for BES, Europe model: SHAP value of each input against its value, coloured by the input with the strongest interaction.

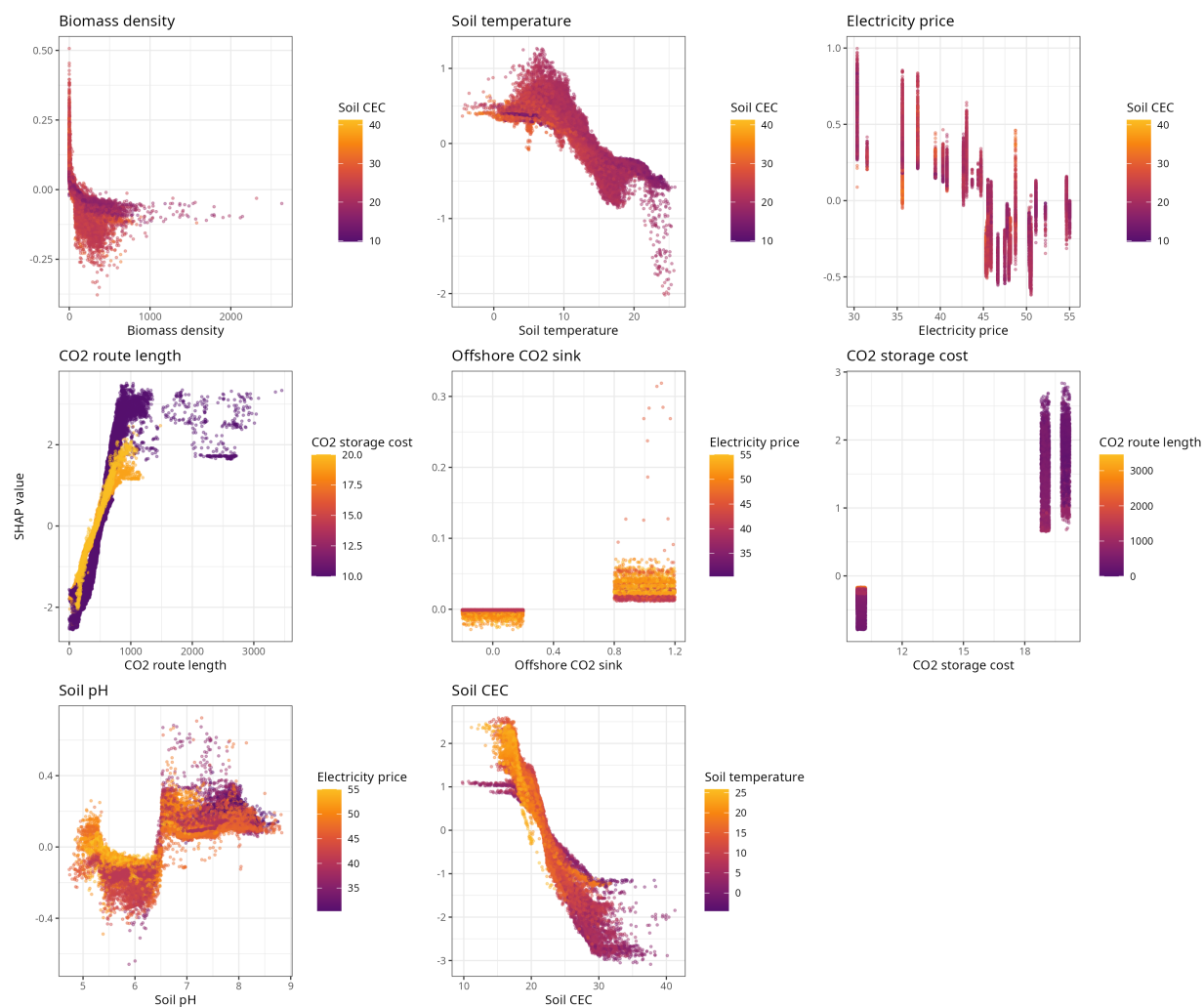


Figure S15: SHAP dependence plots for BEBCS, China model: SHAP value of each input against its value, coloured by the input with the strongest interaction.

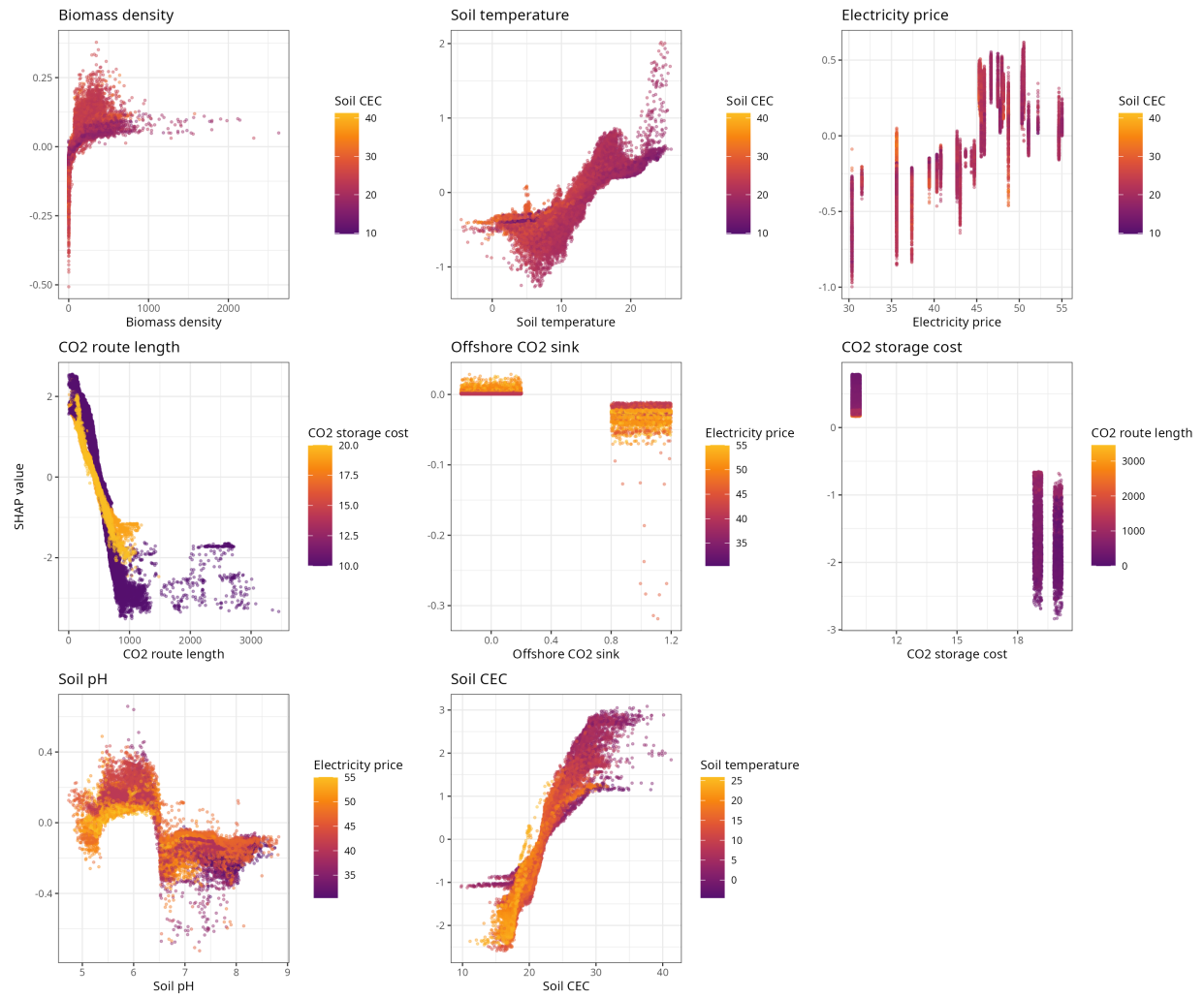


Figure S16: SHAP dependence plots for BECCS, China model: SHAP value of each input against its value, coloured by the input with the strongest interaction.

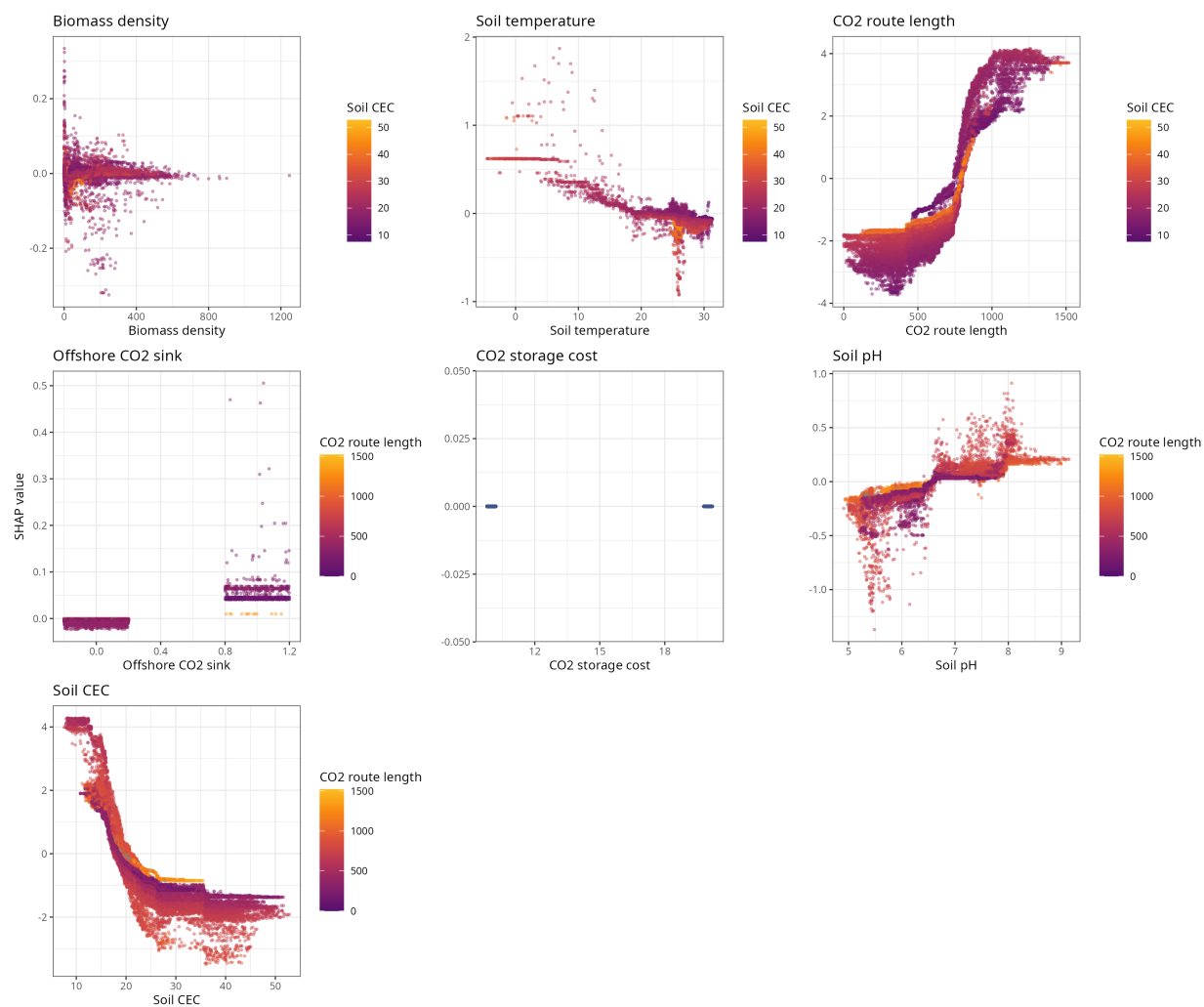


Figure S17: SHAP dependence plots for BEBCS, India model: SHAP value of each input against its value, coloured by the input with the strongest interaction.

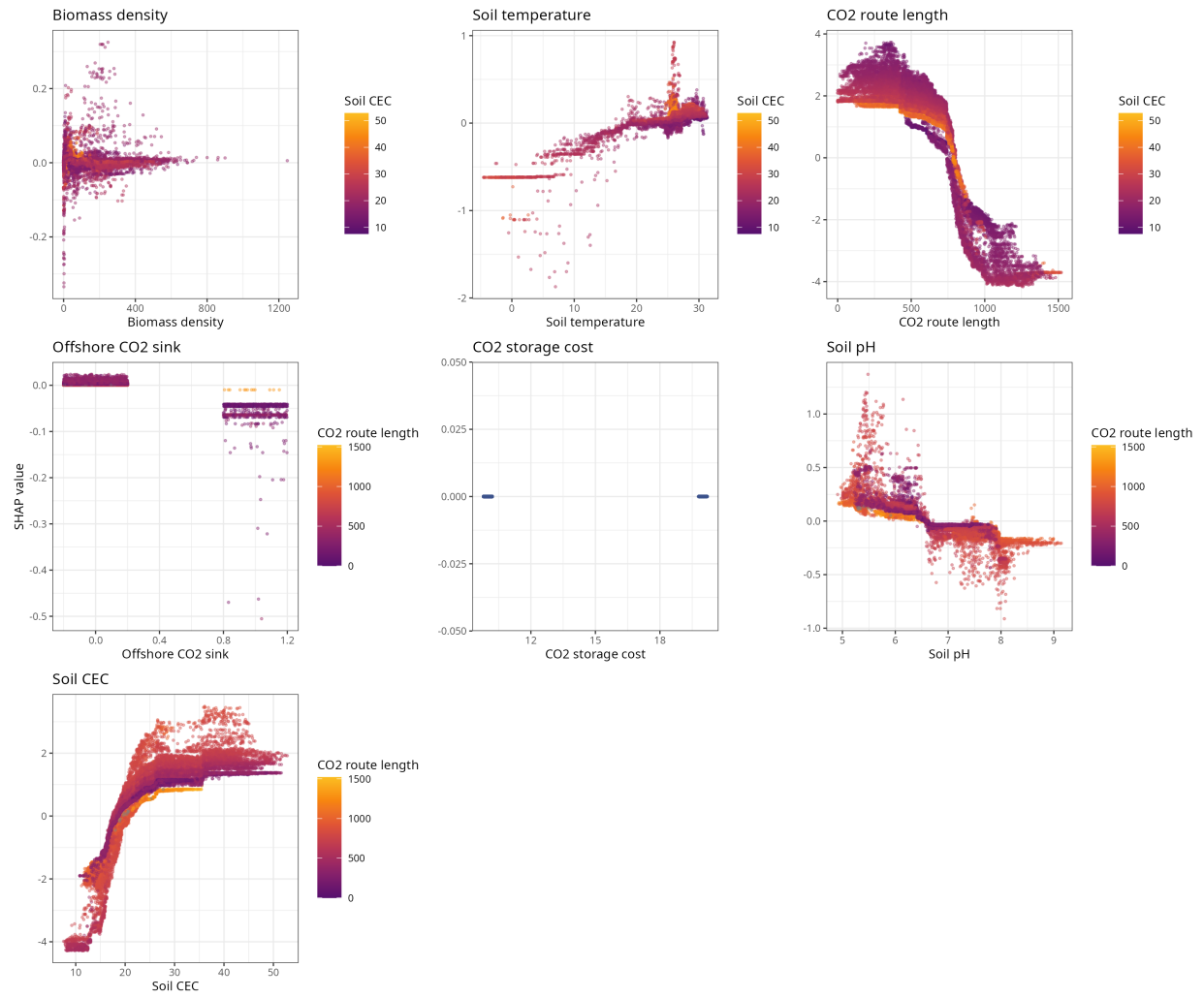


Figure S18: SHAP dependence plots for BECCS, India model: SHAP value of each input against its value, coloured by the input with the strongest interaction.